

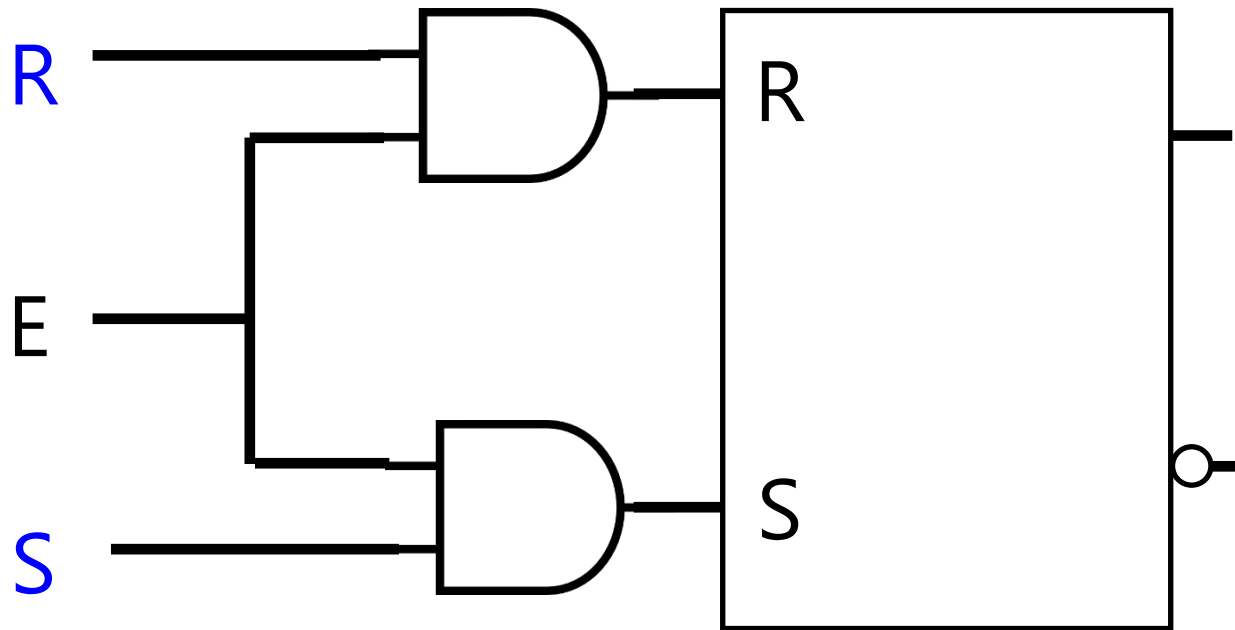


S and R control **how** the
state changes.

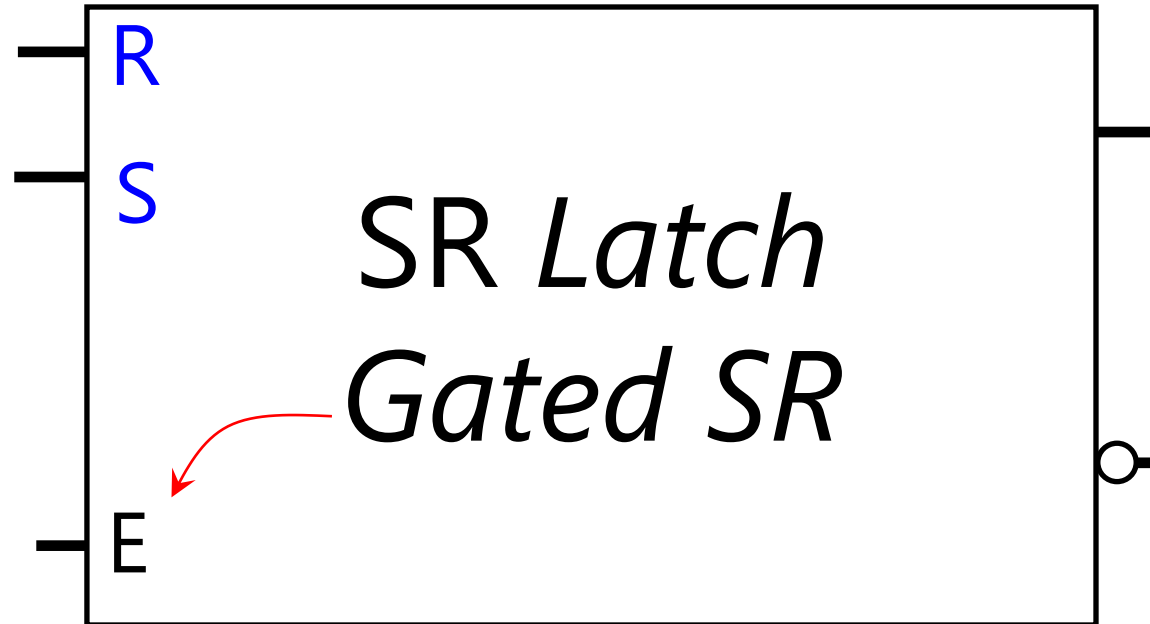
Put a gate/latch on **when** change applies
SR w/ enable input

SR *Latch*

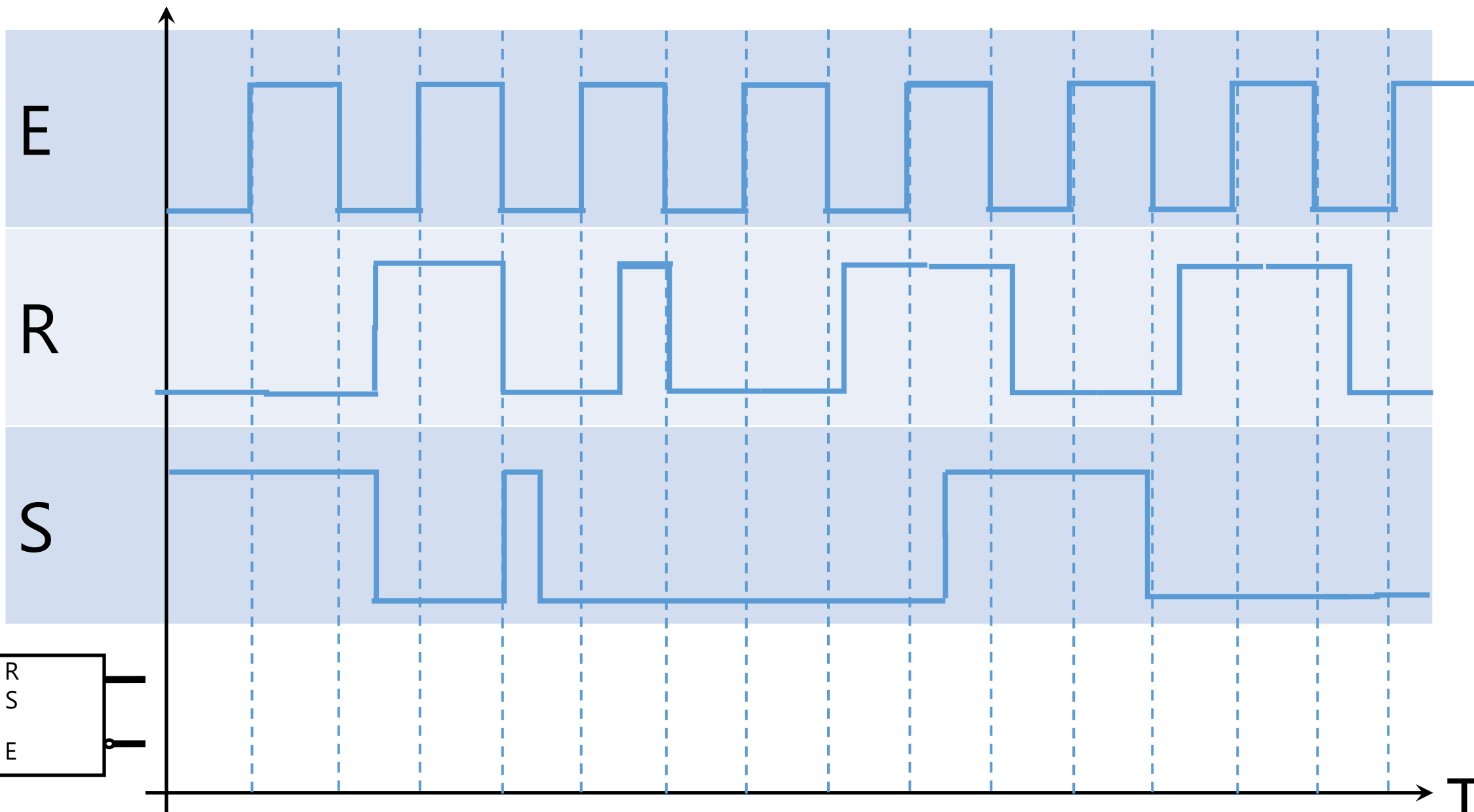
Gated SR



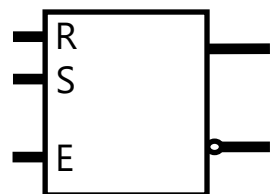
E	S	R	S	R	Q	Q'
0	x	x	0	0	Q_t	Q'_t
1	0	0	0	0	Q_t	Q'_t
1	0	1	0	1	0	1
1	1	0	1	0	1	0
1	1	1	1	1	x	x



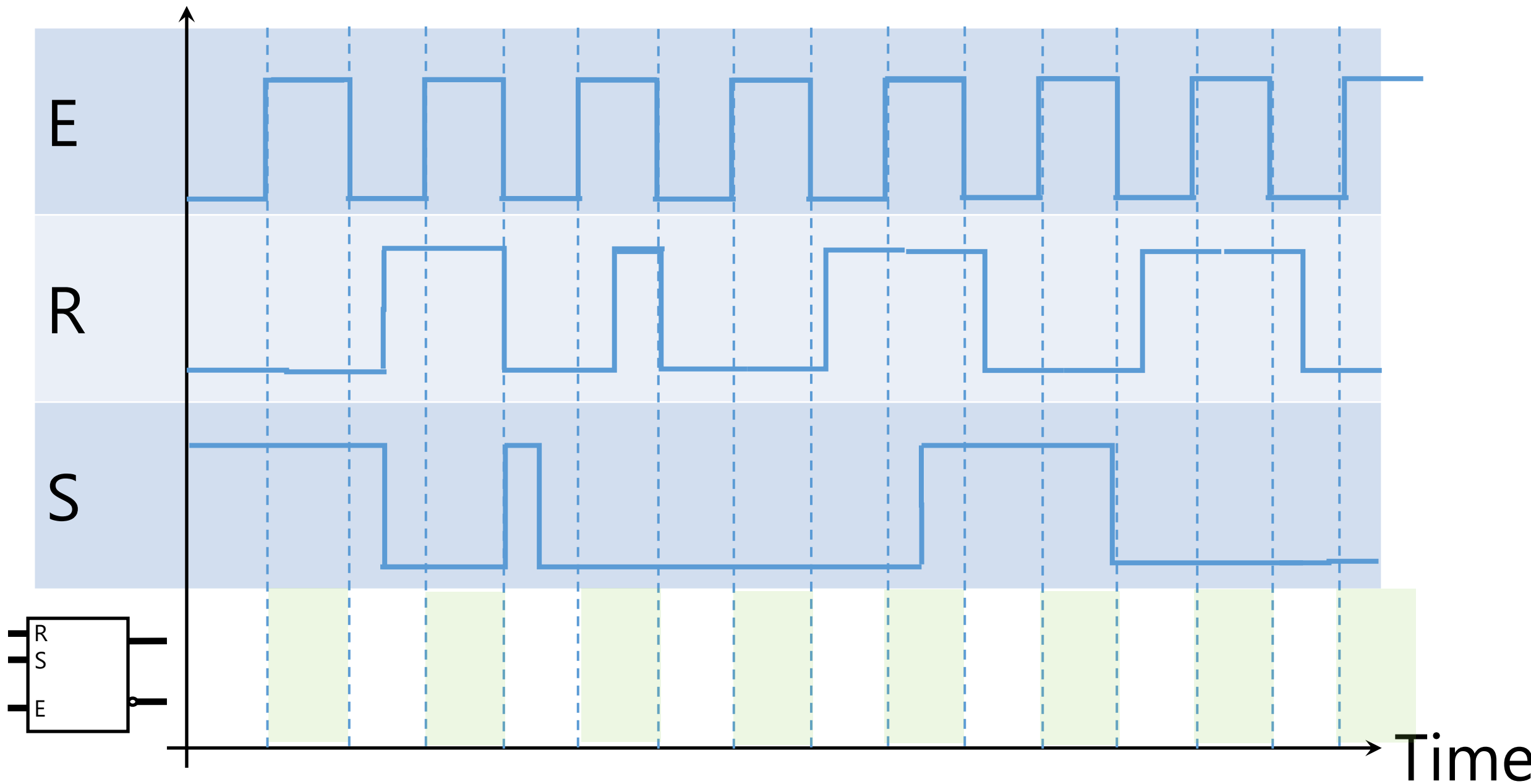
Voltage



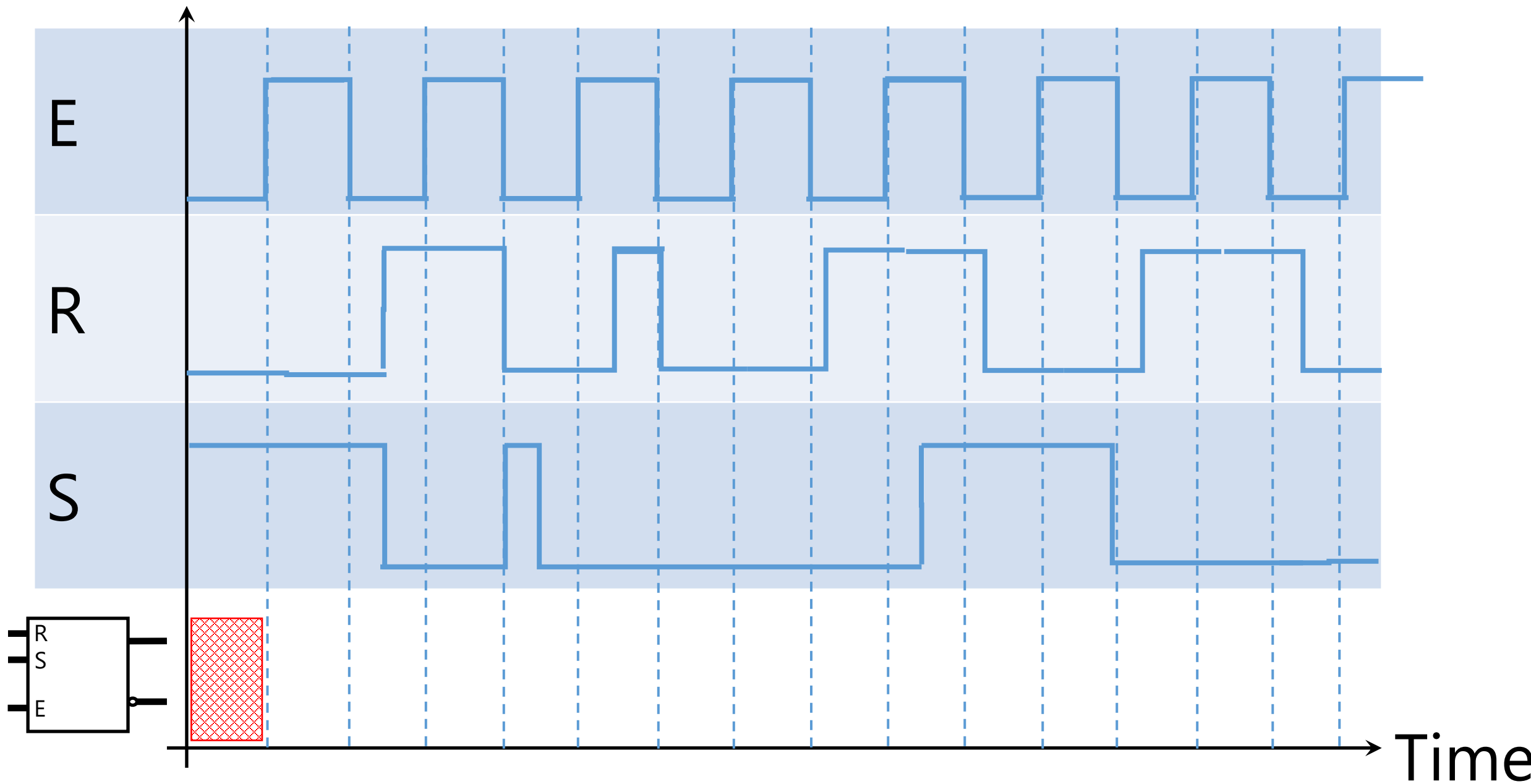
Time



Voltage

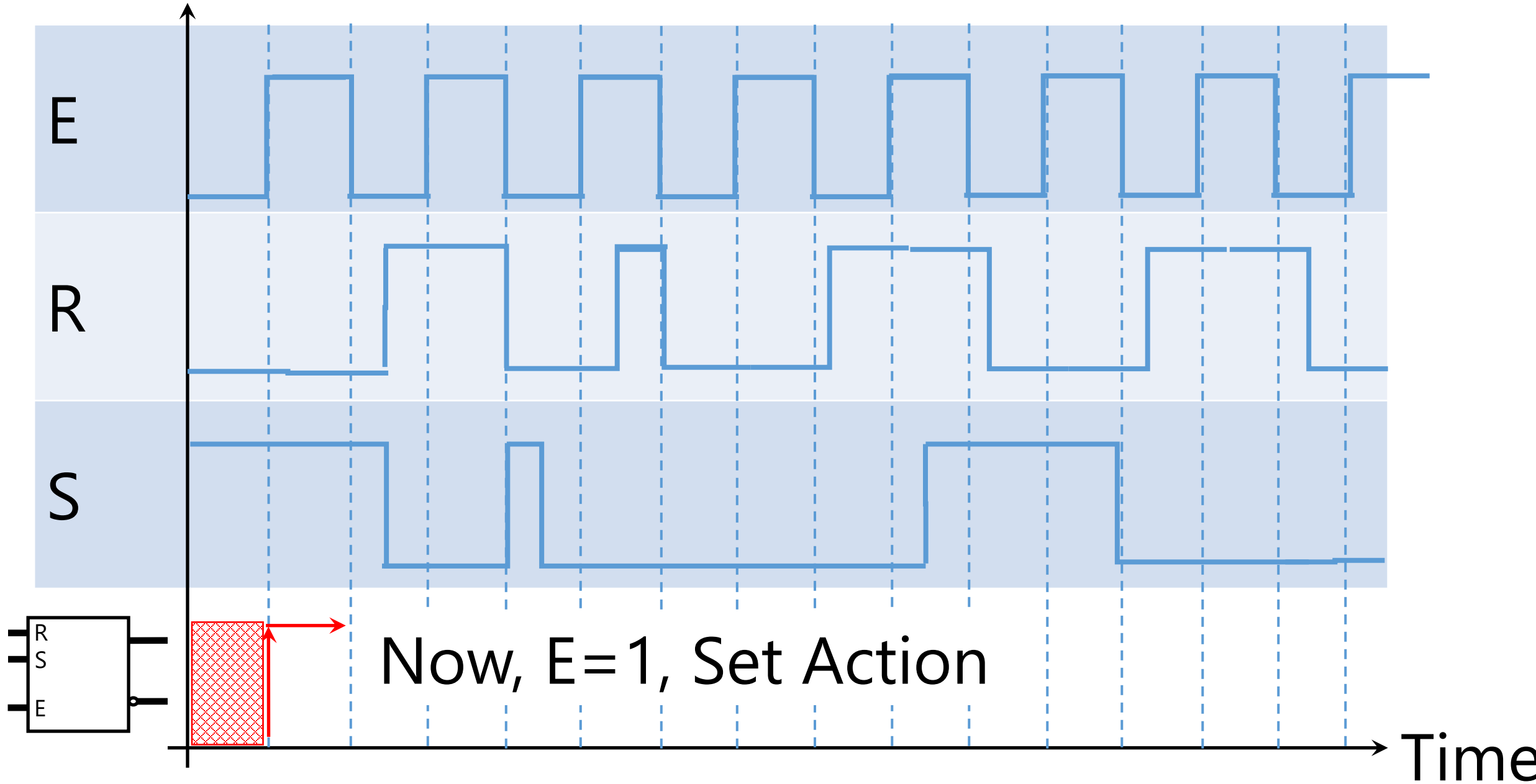


Voltage

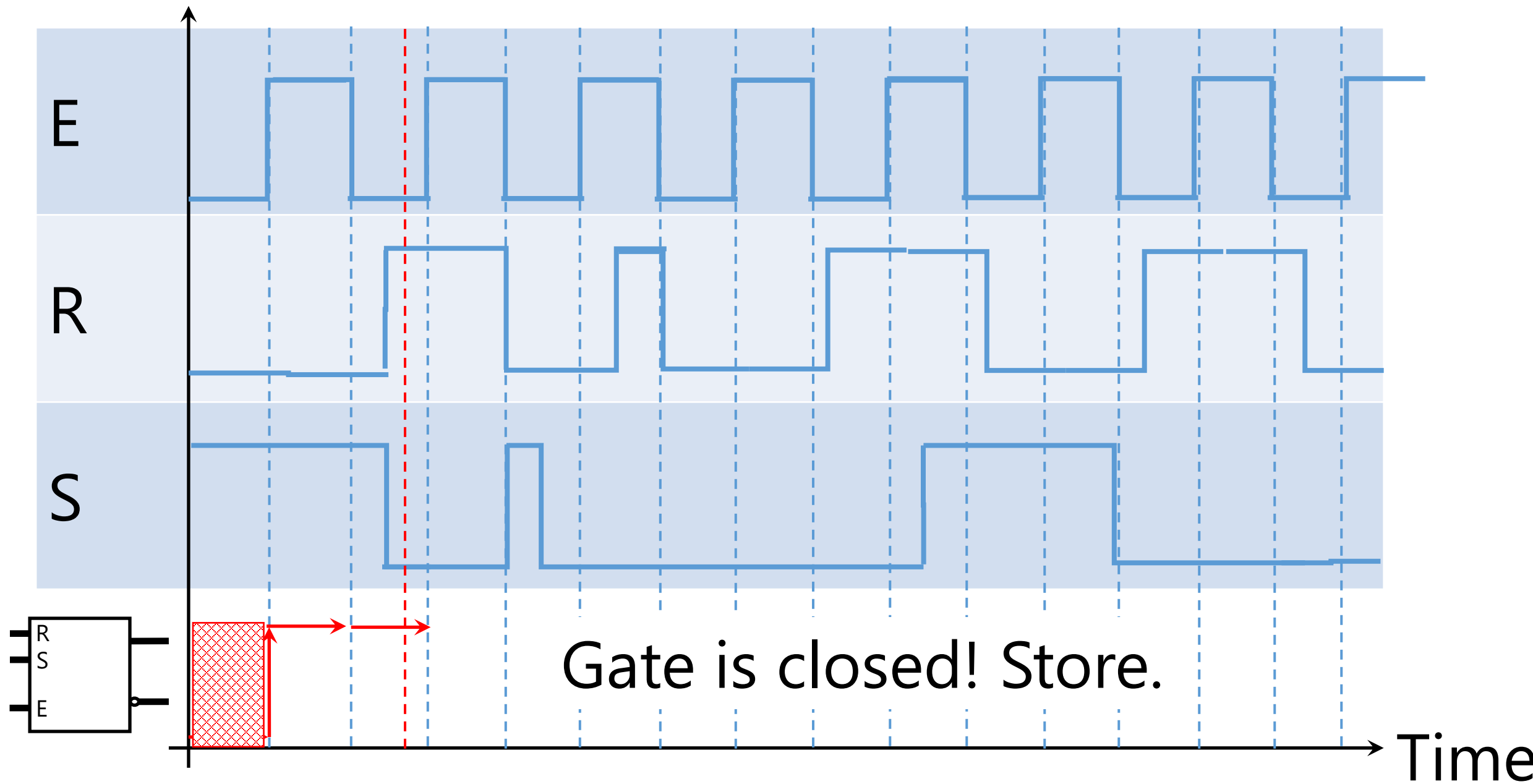


Time

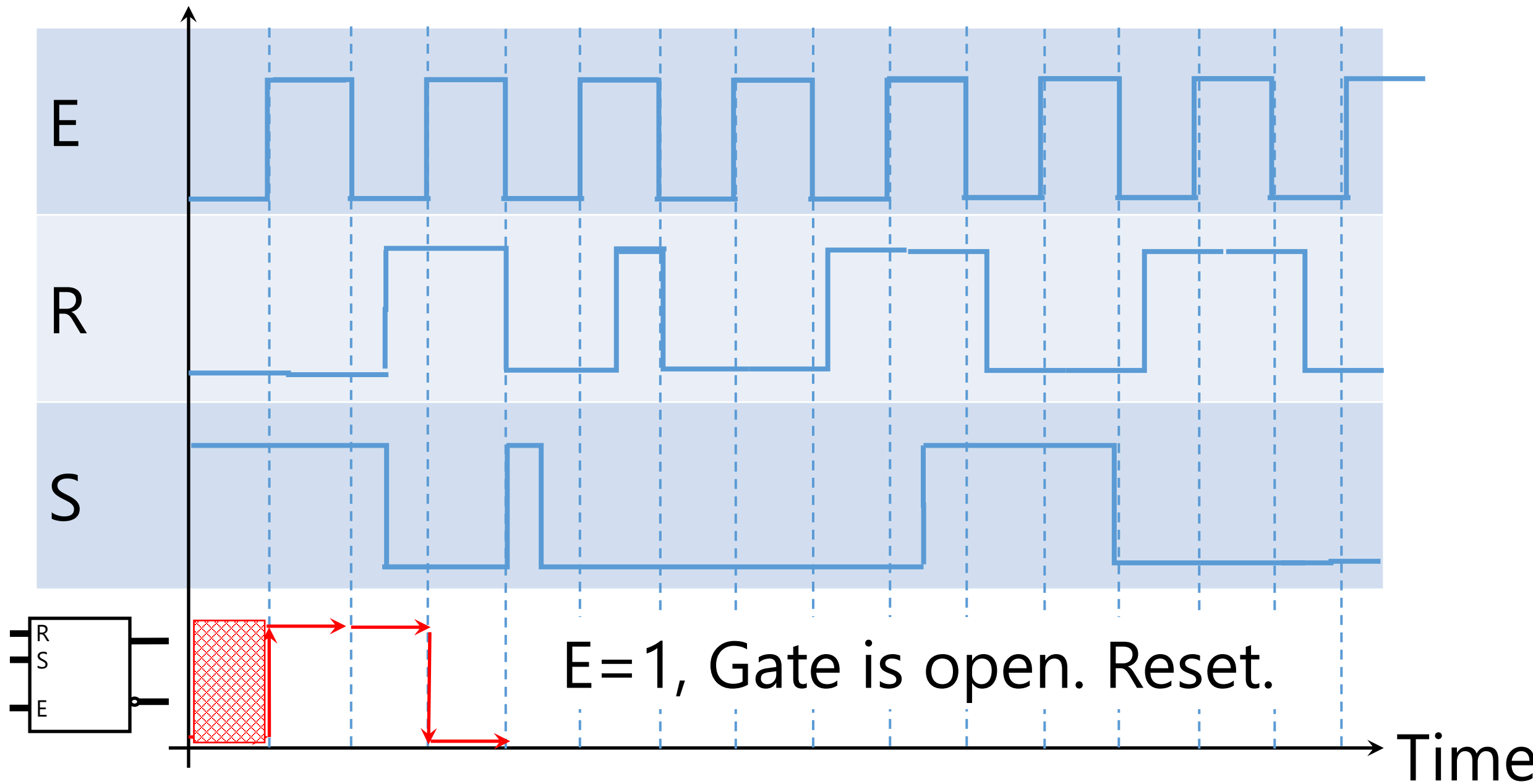
Voltage



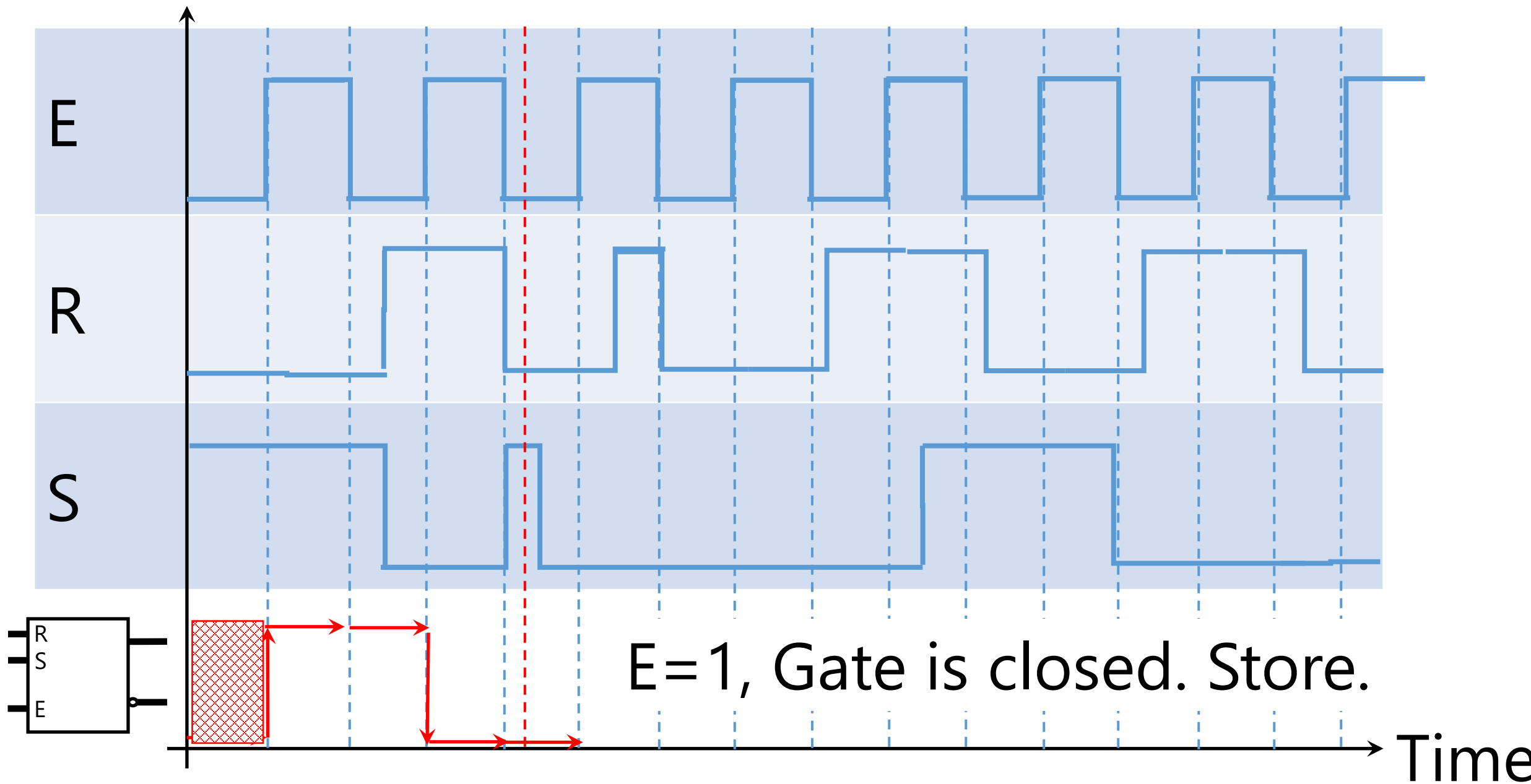
Voltage



Voltage

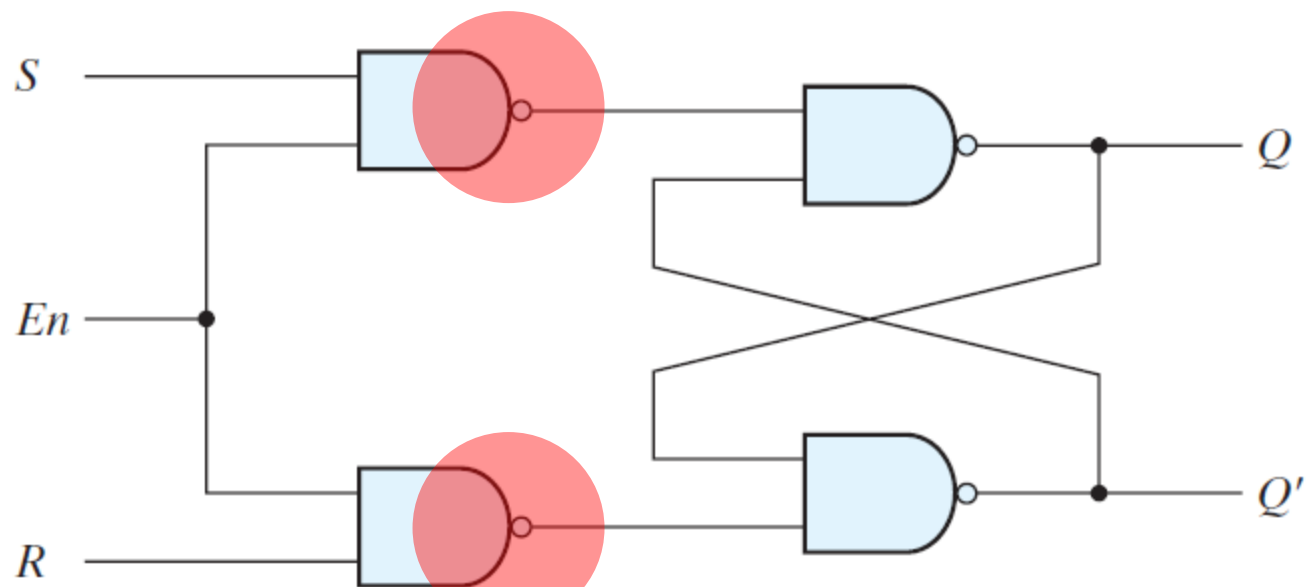


Voltage



SR *Latch*

NAND

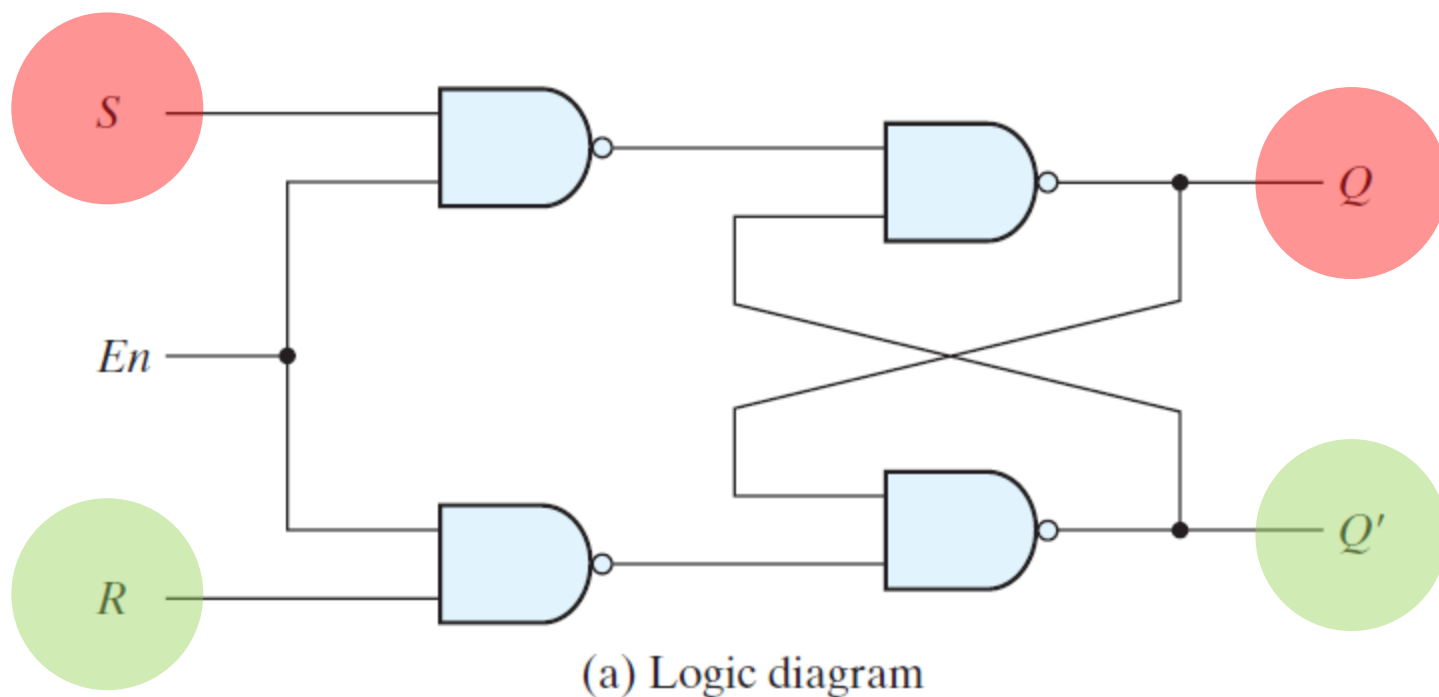


(a) Logic diagram

En	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

FIGURE 5.5
SR latch with control input

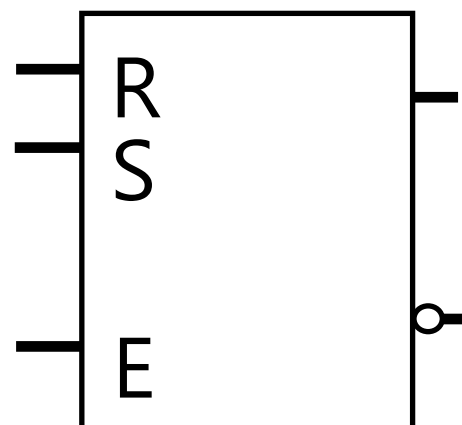


En	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

FIGURE 5.5
SR latch with control input

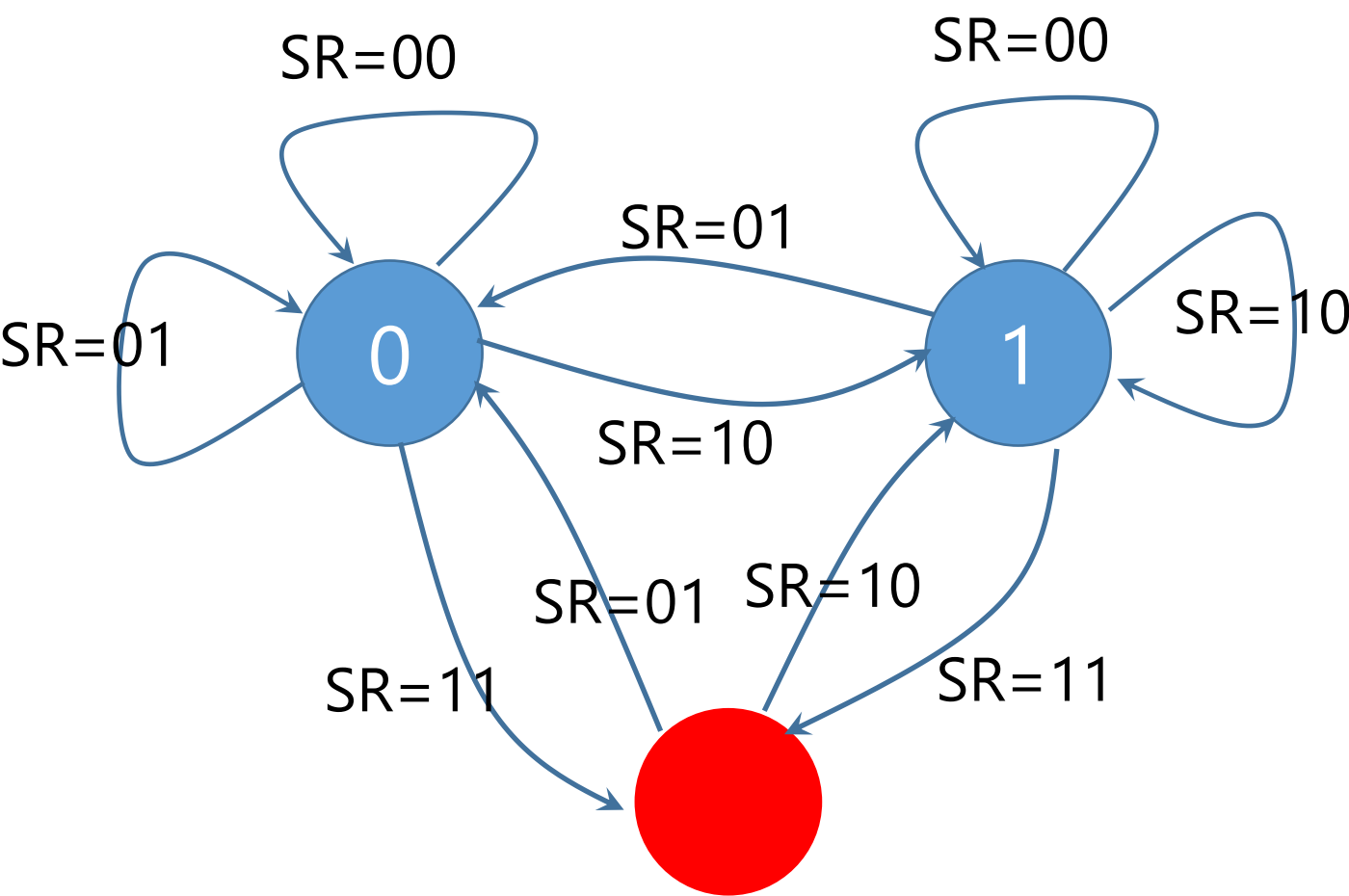
Block Diagram



Characteristic Table

S	R	Q
0	0	Q_t
0	1	0
1	0	1
1	1	$\times$

State Transition Diagram



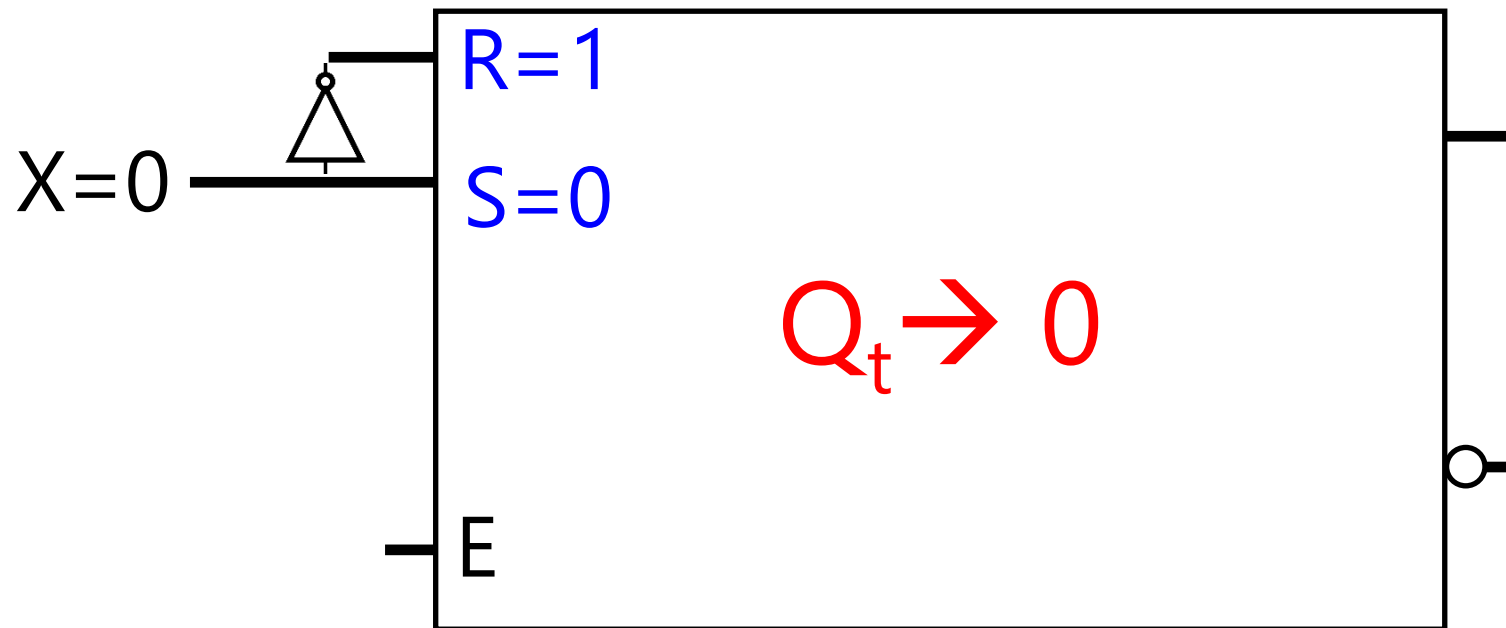
Other *Latches*



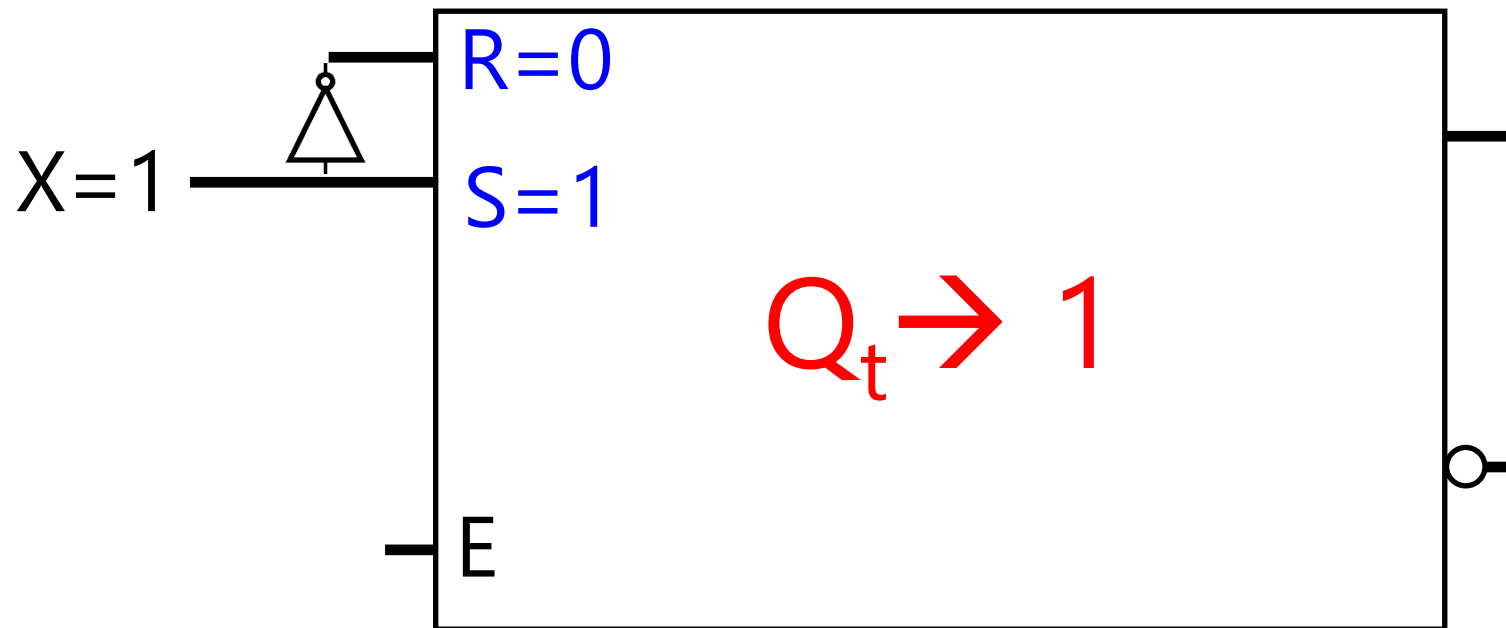
Let's avoid the forbidden action! $R \neq S$
 $R = S'$ and $S = R'$



X	E	S	R	Q	Q'
0	0	x	x	Q_t	Q'_t
1	0	x	x	Q_t	Q'_t
0	1	0	1	0	1
1	1	1	0	1	0
Never happens		1	1	$\times$	$\times$



X	E	S	R	Q	Q'
0	0	$\times$	$\times$	Q_t	Q'_t
1	0	$\times$	$\times$	Q_t	Q'_t
0	1	0	1	0	1
1	1	1	0	1	0
Never happens		1	1	$\times$	$\times$



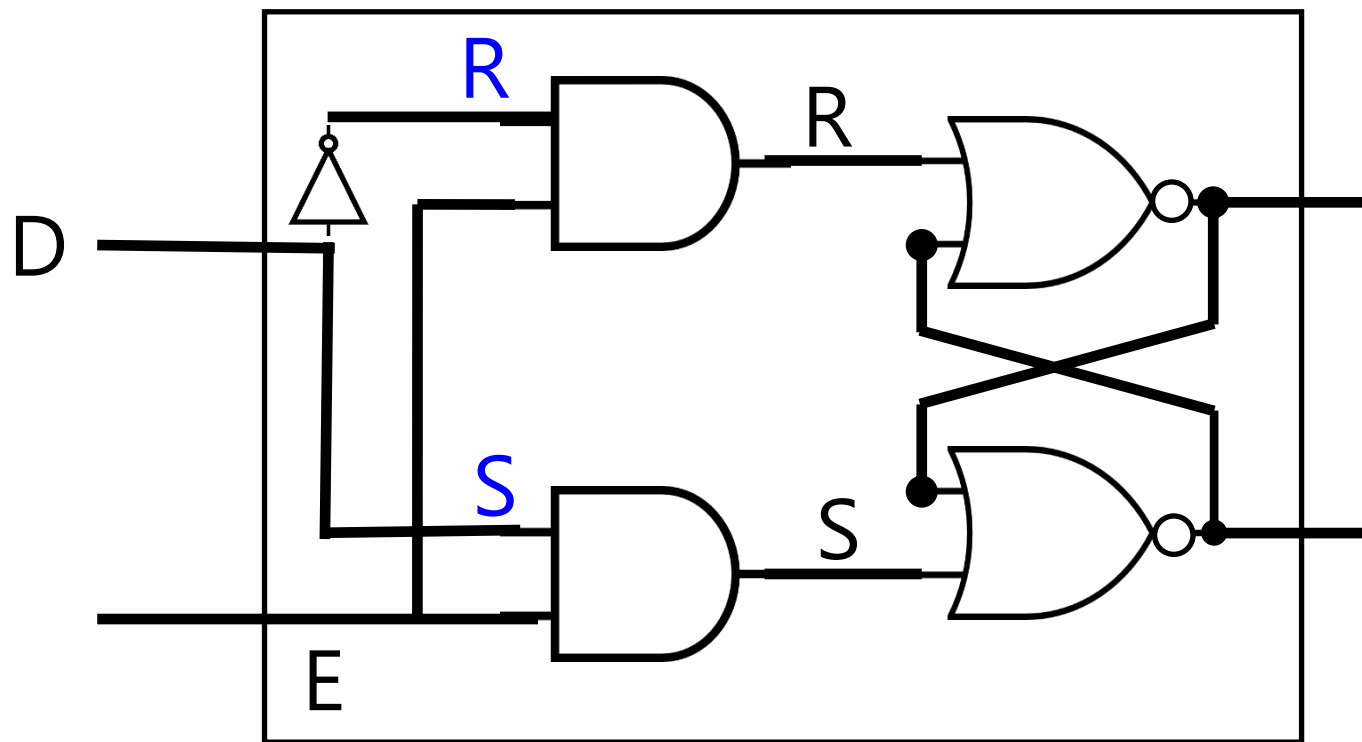
X	E	S	R	Q	Q'
0	0	x	x	Q_t	Q'_t
1	0	x	x	Q_t	Q'_t
0	1	0	1	0	1
1	1	1	0	1	0
Never happens		1	1	$\times$	$\times$



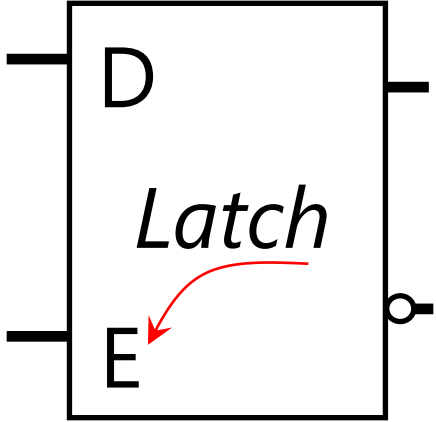
D	E	Q
0	0	Q_t
1	0	Q_t
0	1	0
1	1	1



D	Q
0	0
1	1



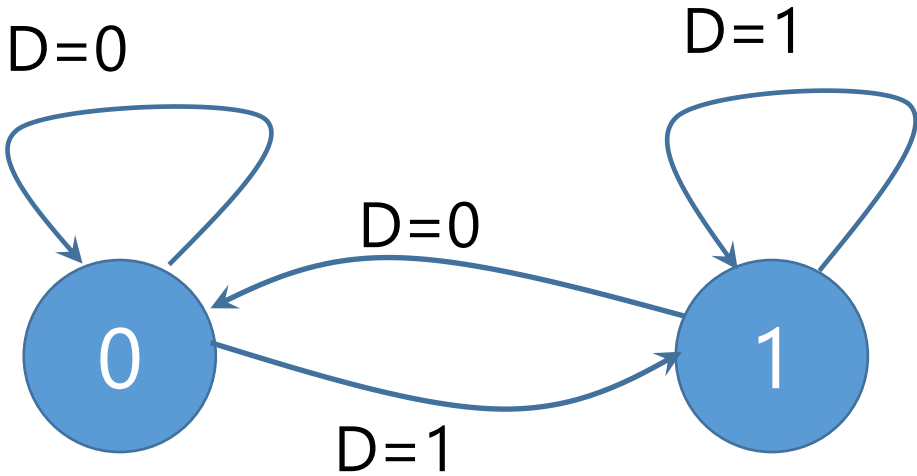
Block Diagram



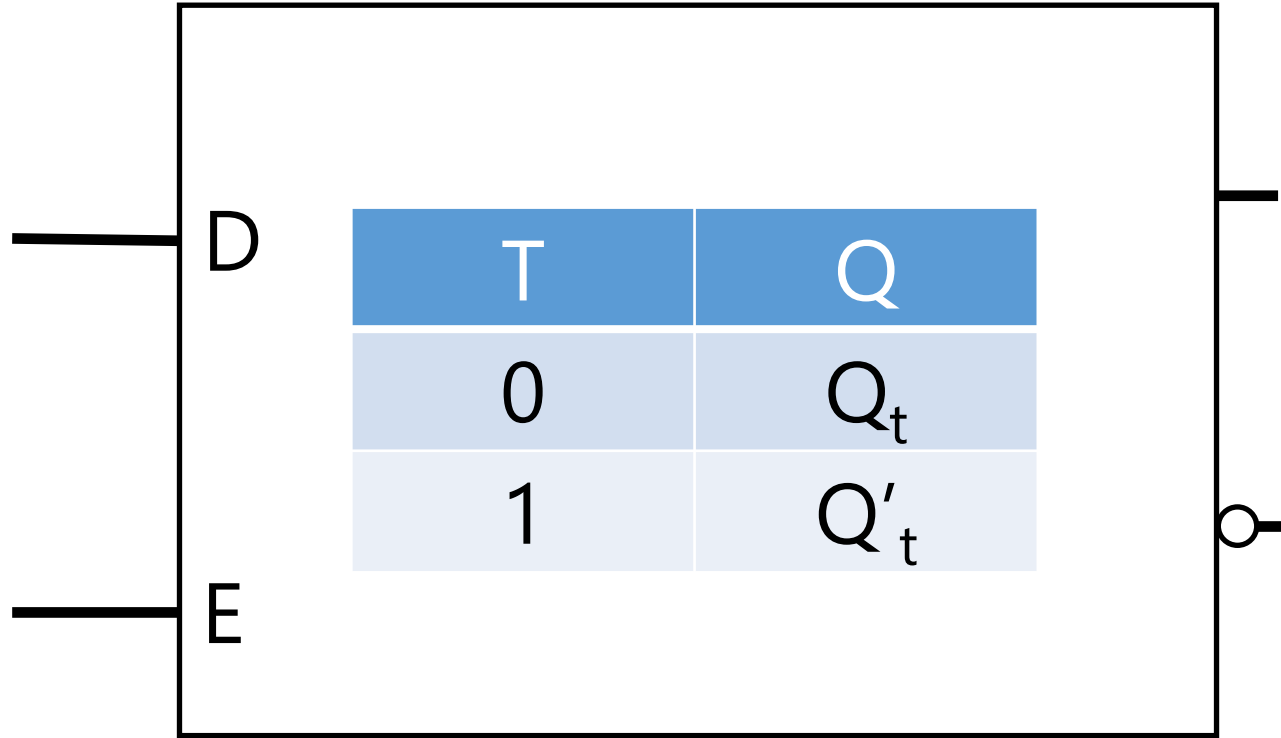
Characteristic Table

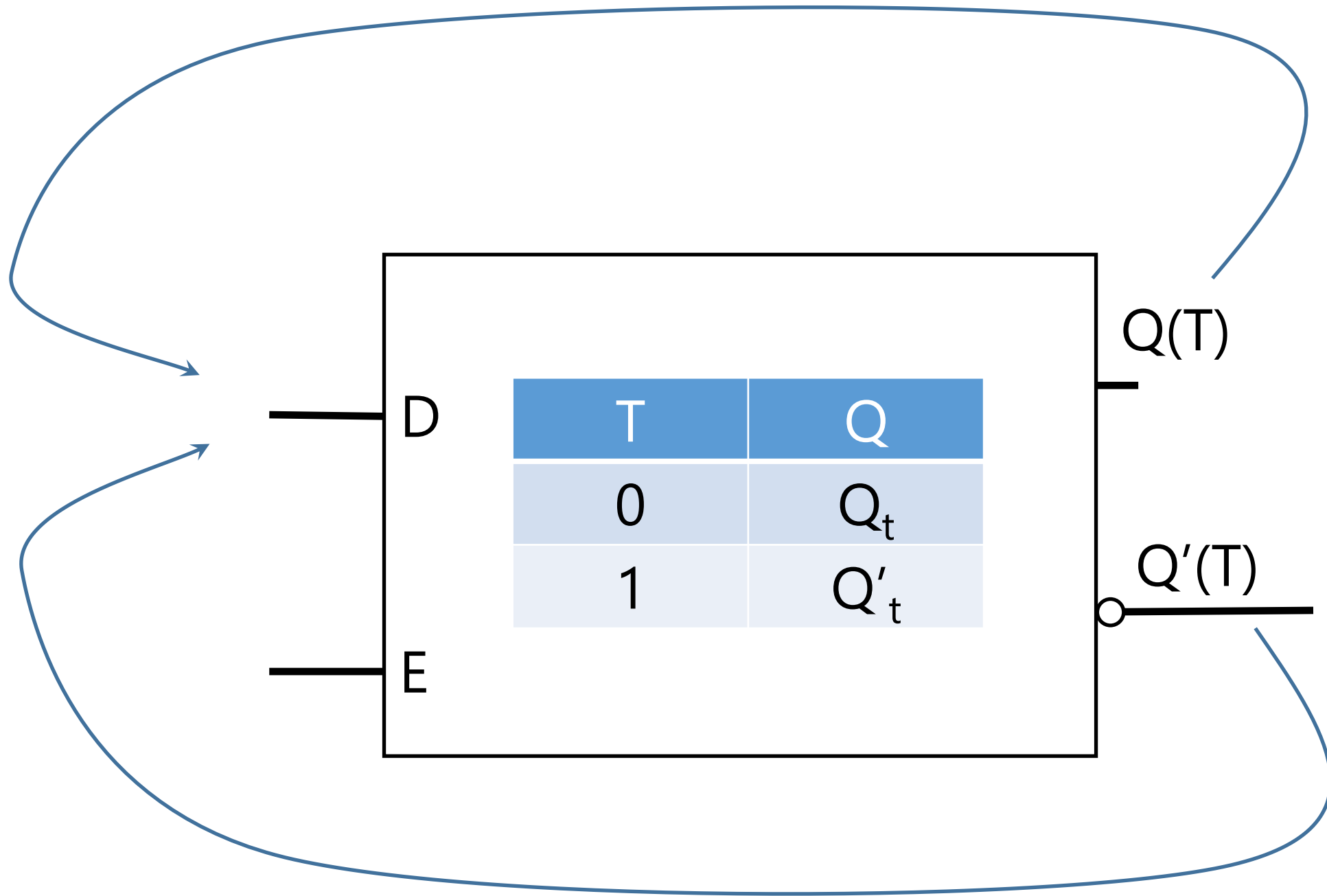
D	Q
0	0
1	1

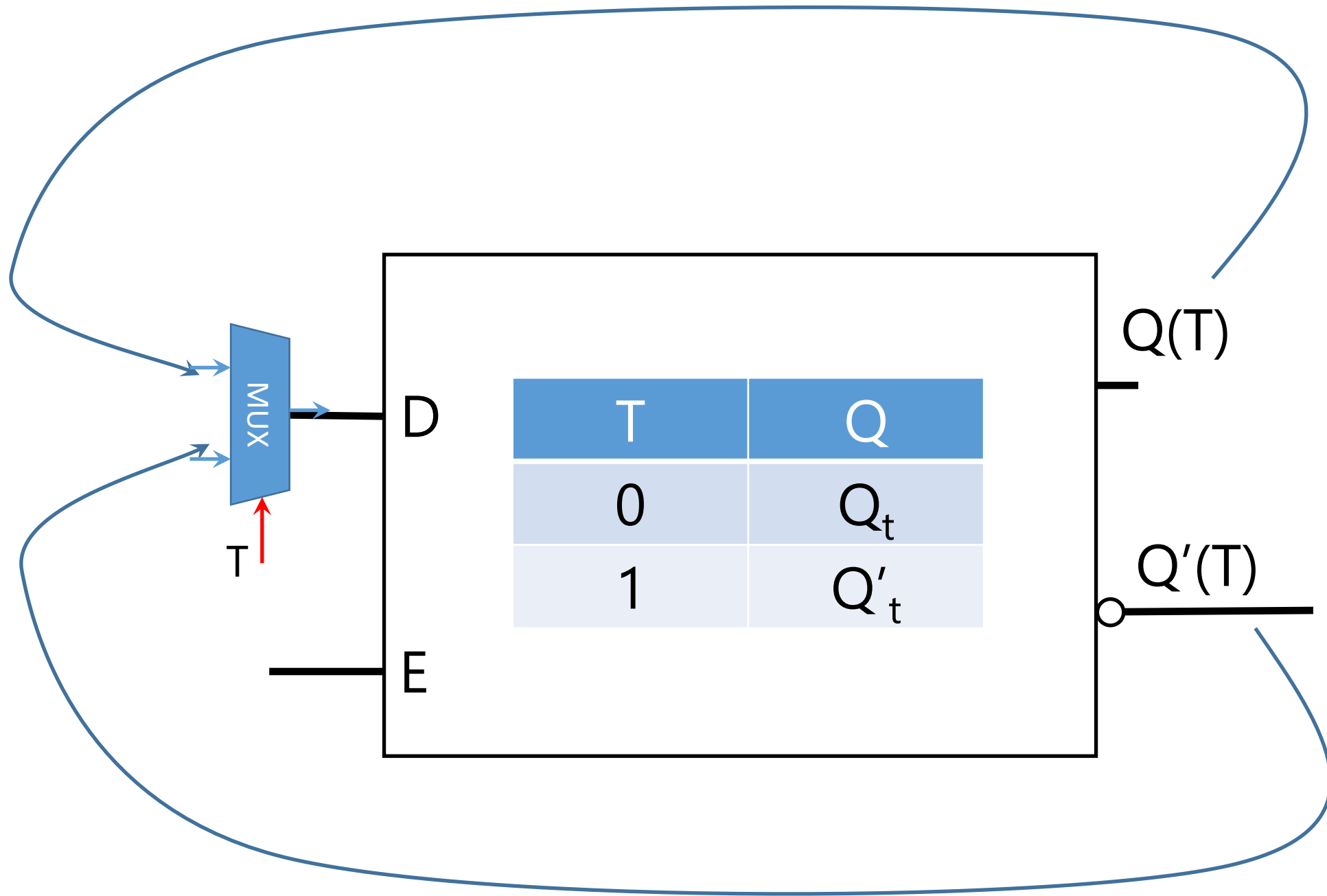
State Transition Diagram

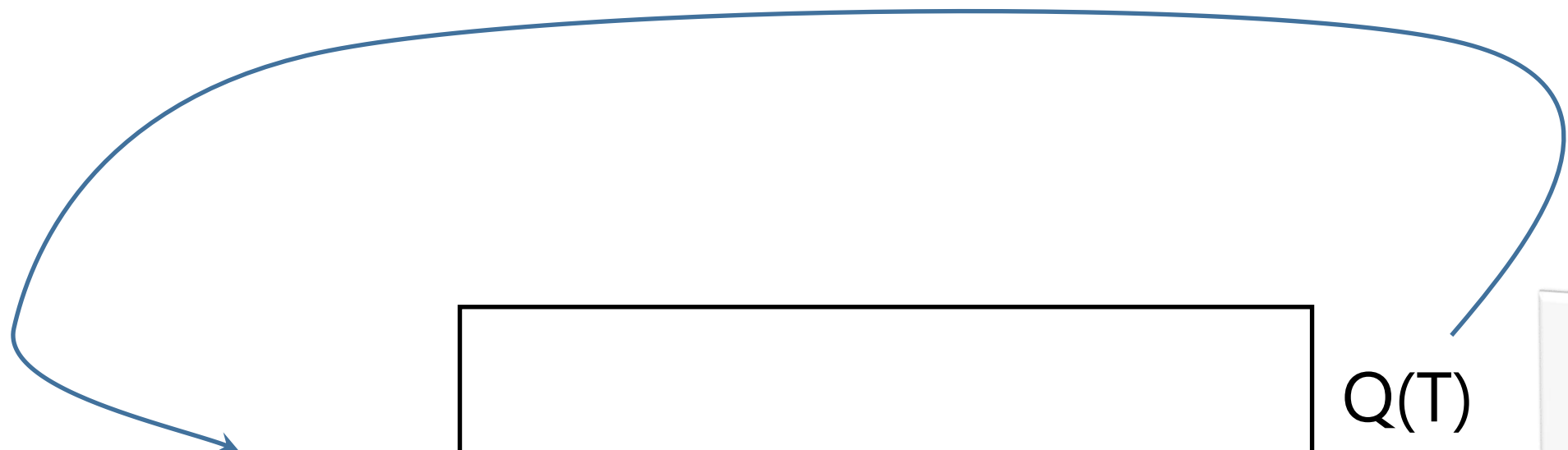


T *Latch*

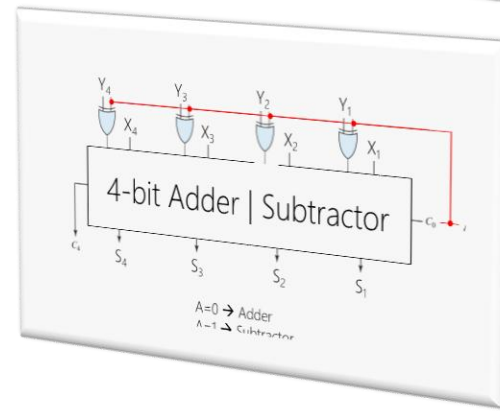
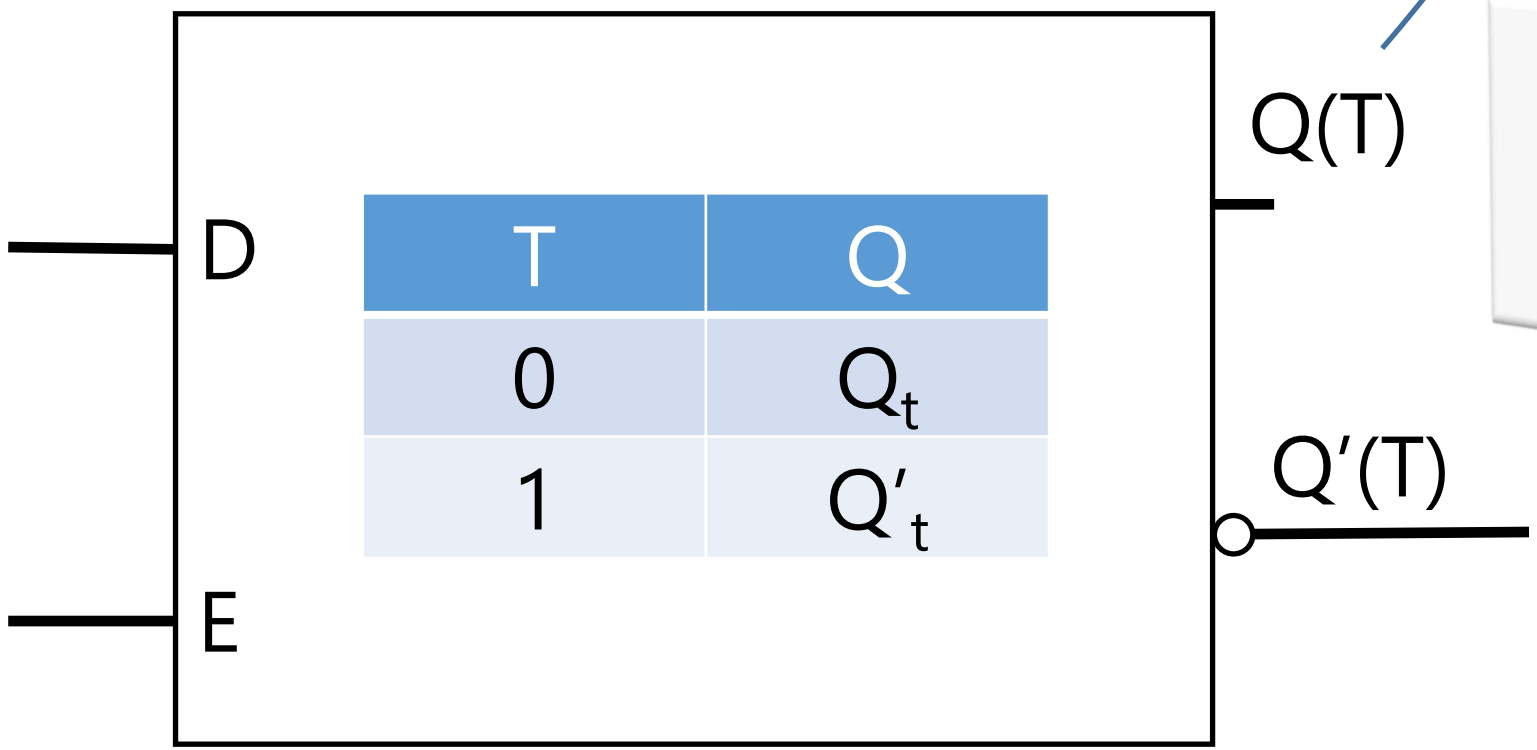


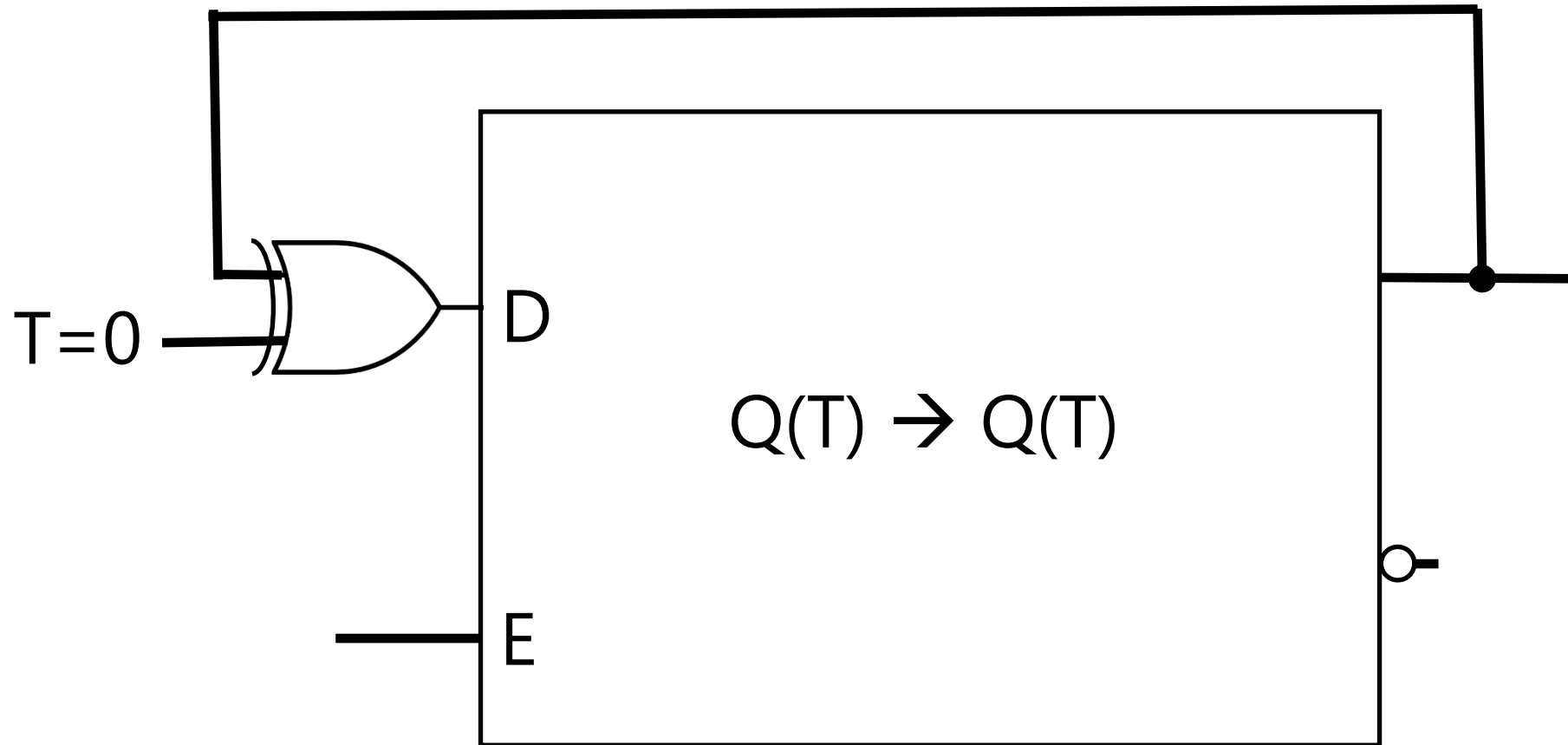


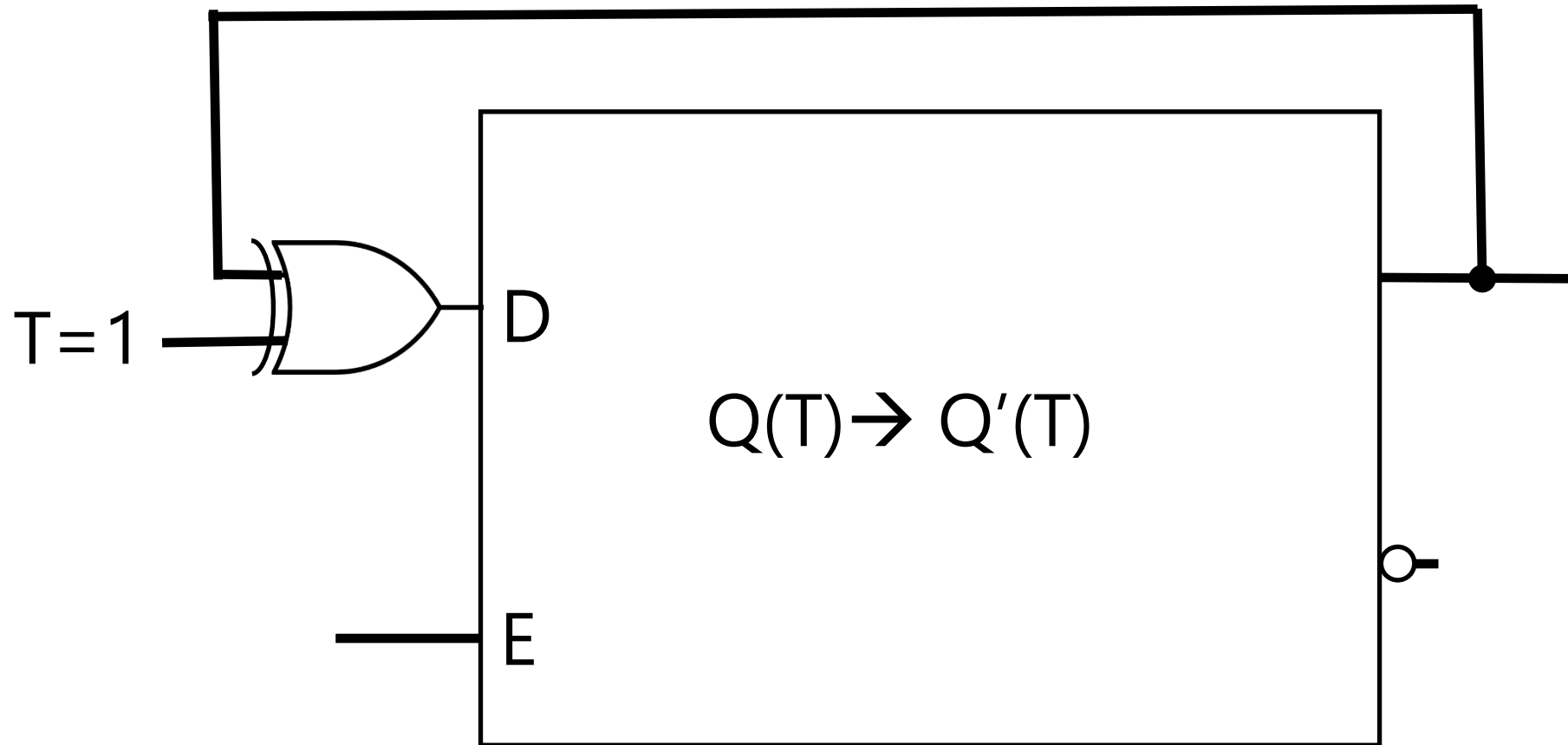


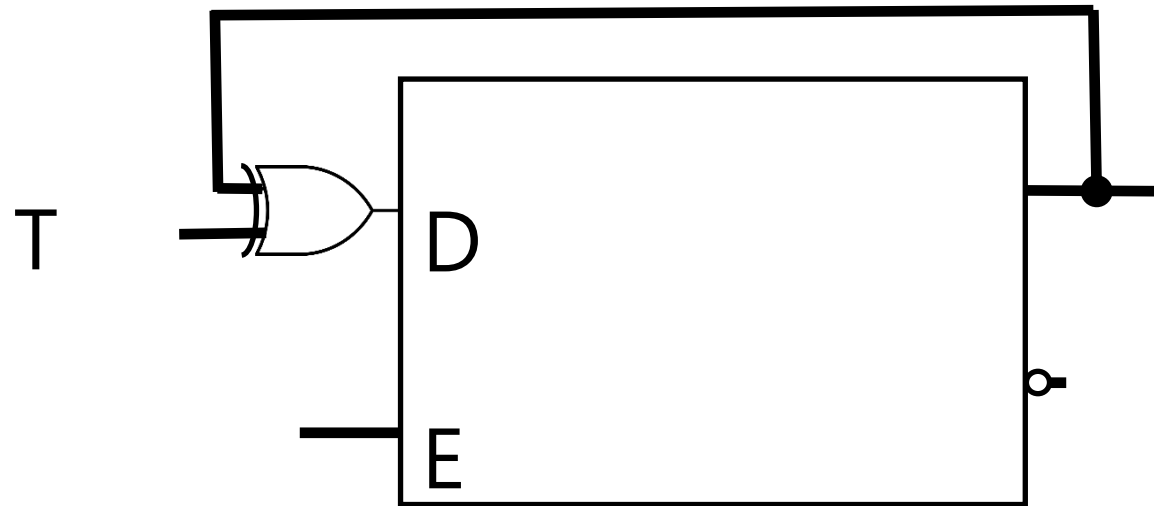


$$C_0 ? 0 = C_0$$
$$C_0 ? 1 = C'_0$$

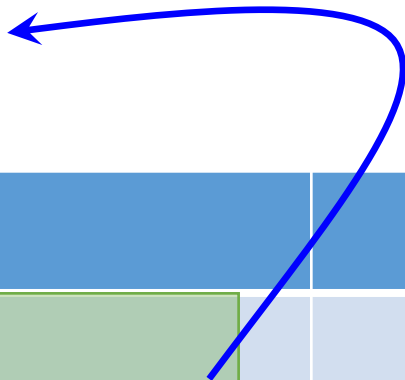




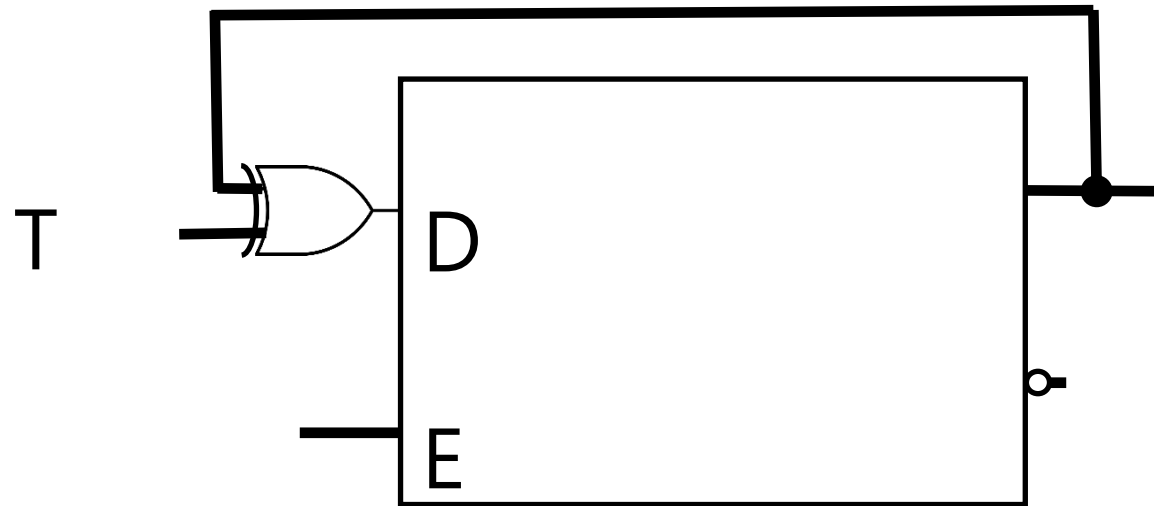




Store

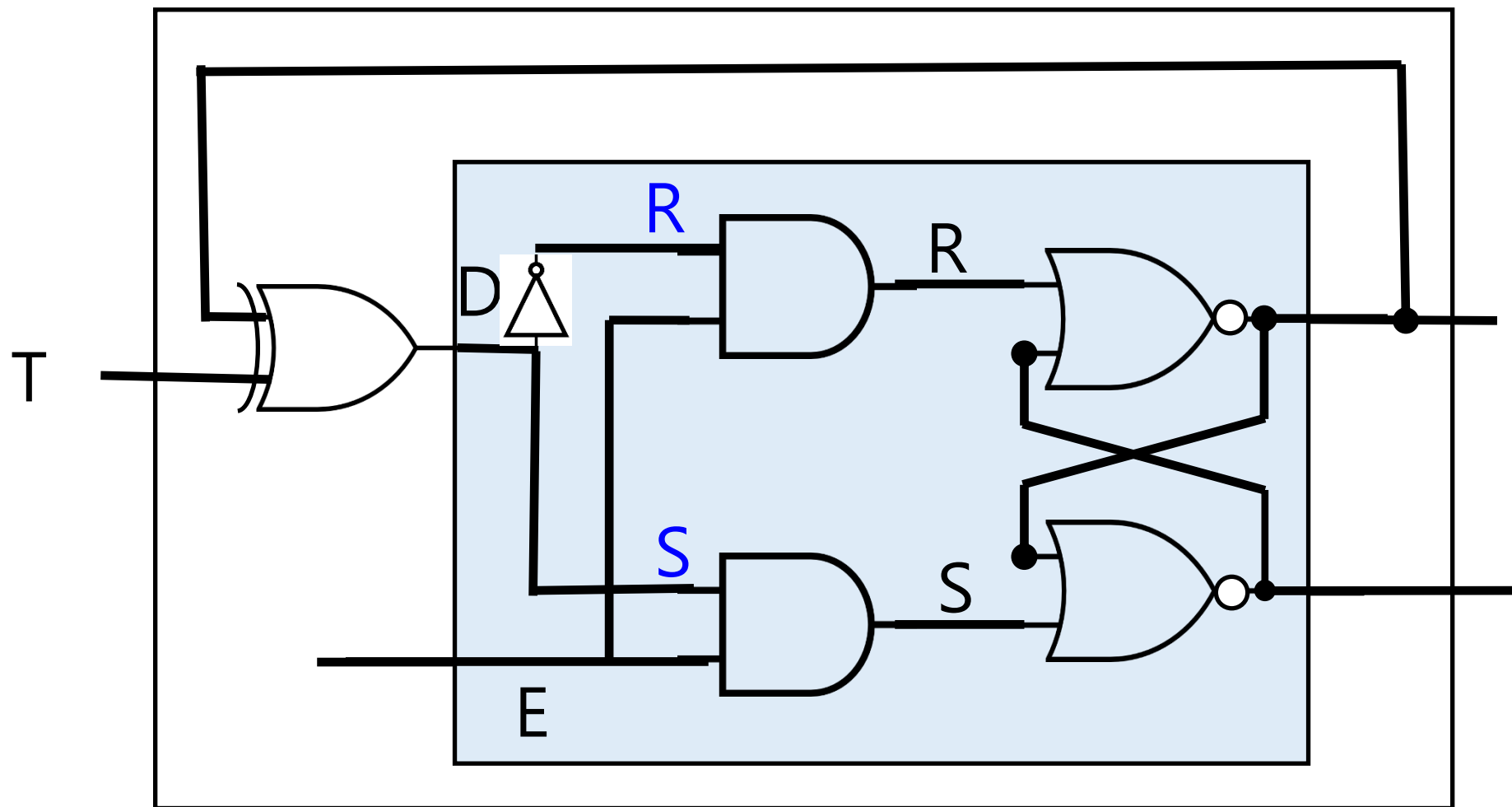


T	D	Q
0	$Q \oplus 0$	Q
1	$Q \oplus 1$	Q'

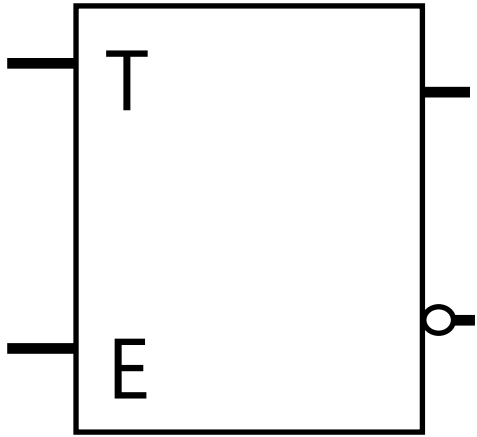


Complement

T	D	Q
0	$Q \oplus 0$	Q
1	$Q \oplus 1$	Q'



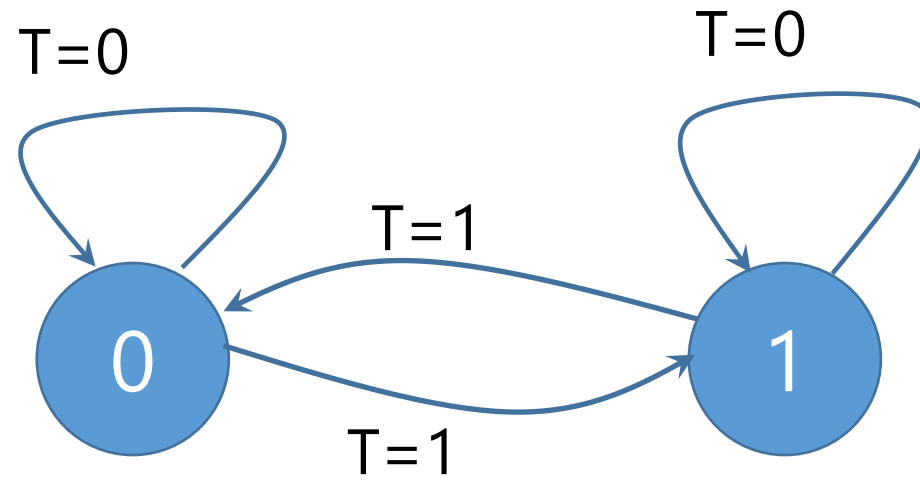
Block Diagram



Characteristic Table

T	Q
0	Q_t
1	Q'_t

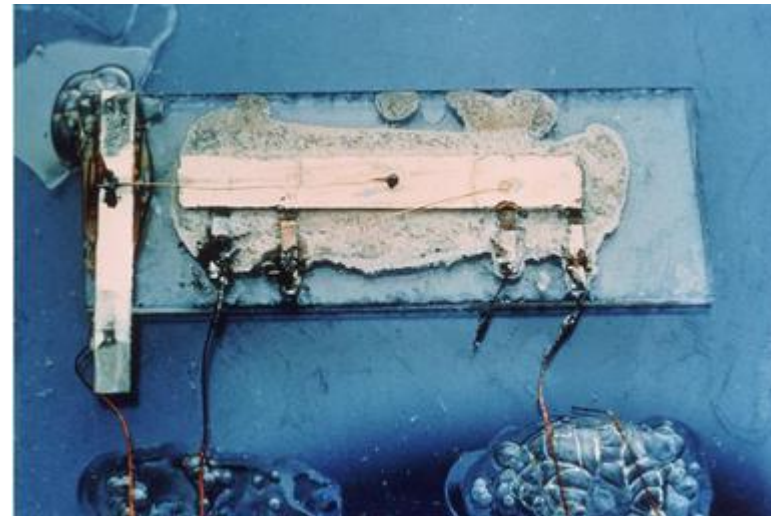
State Transition Diagram

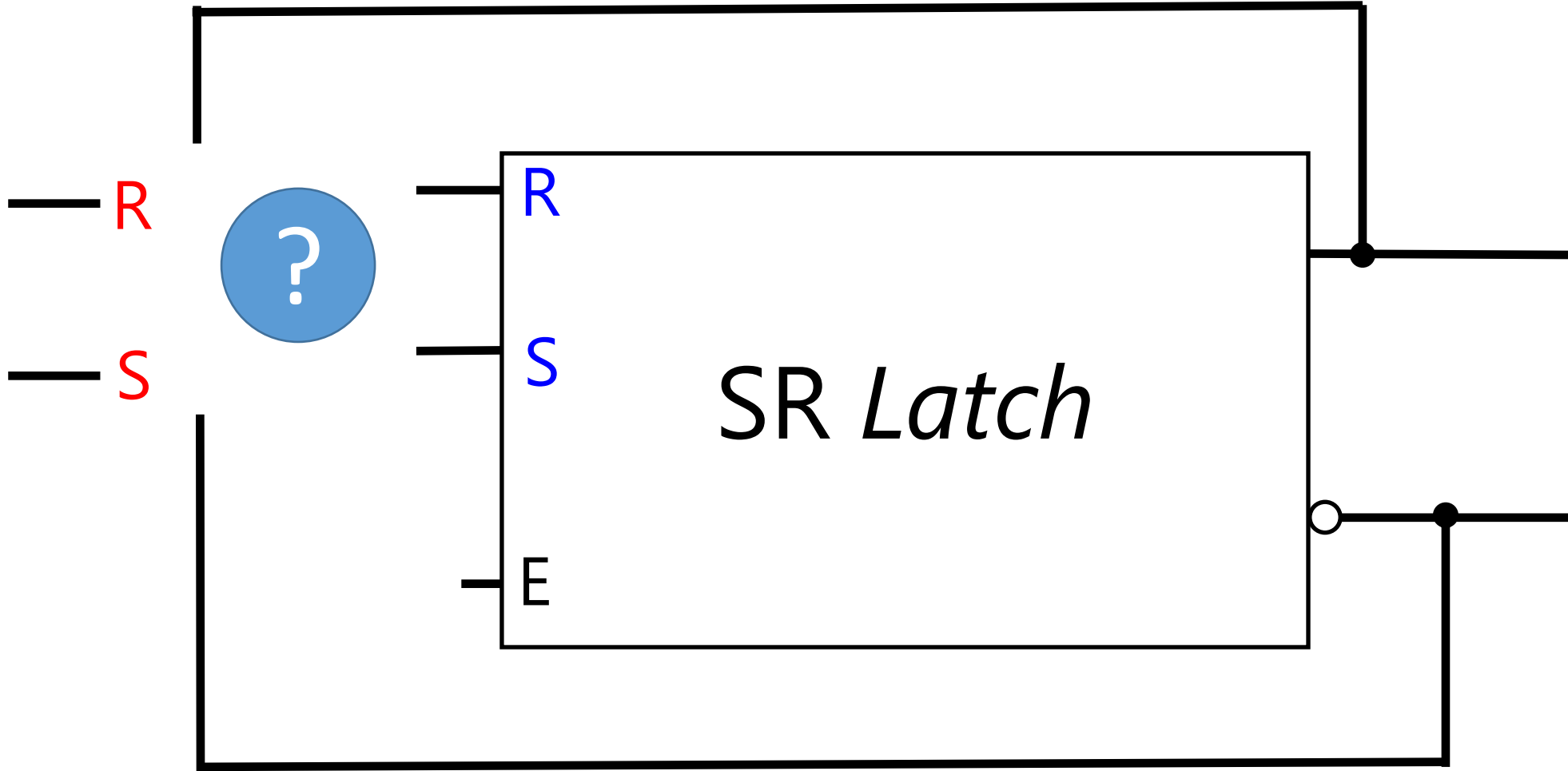


JK Latch



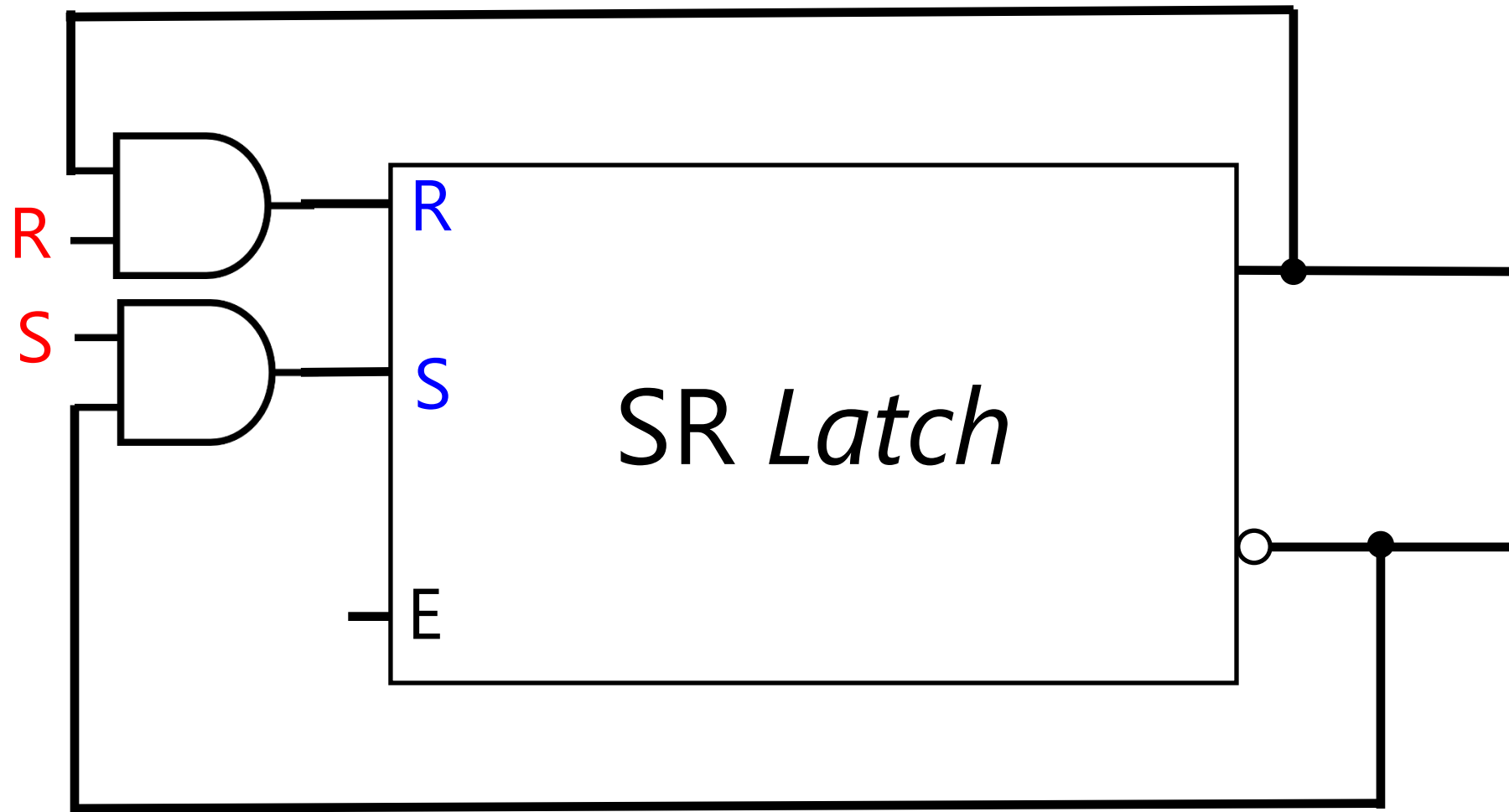
Jack St. Clair Kilby
(Nov. 8, 1923 – June 20, 2005)
Electrical Engineer
The 1st integrated circuit
1958
Nobel Prize in Physics, 2000

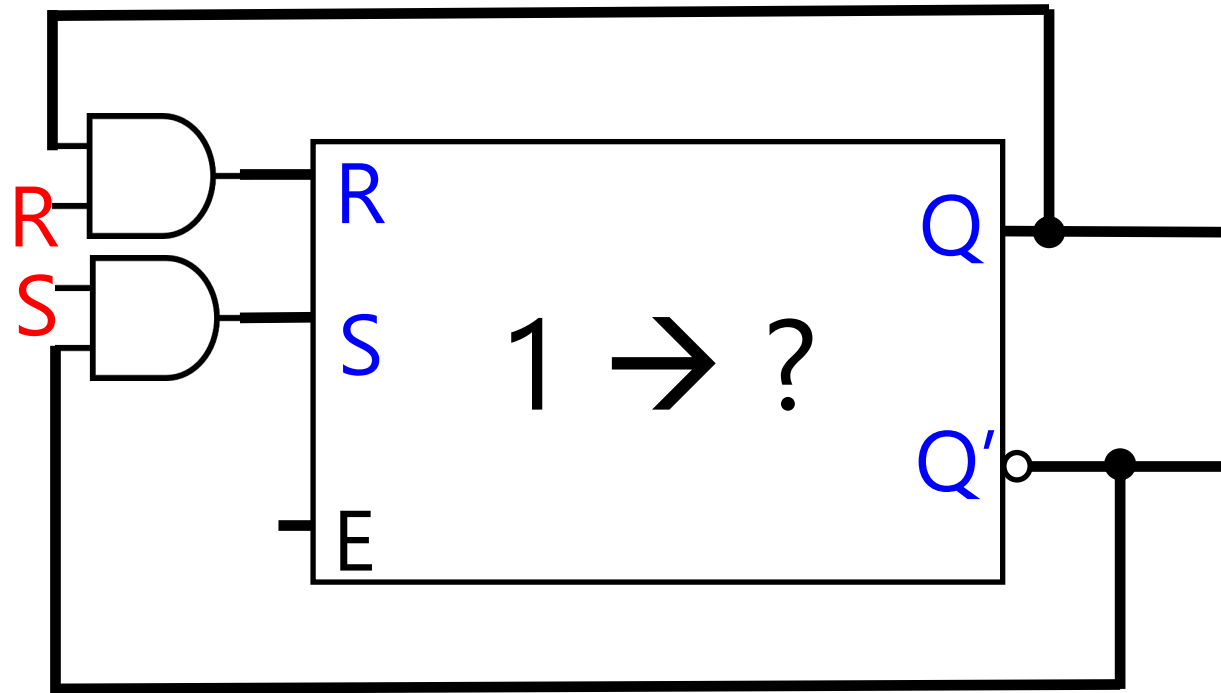




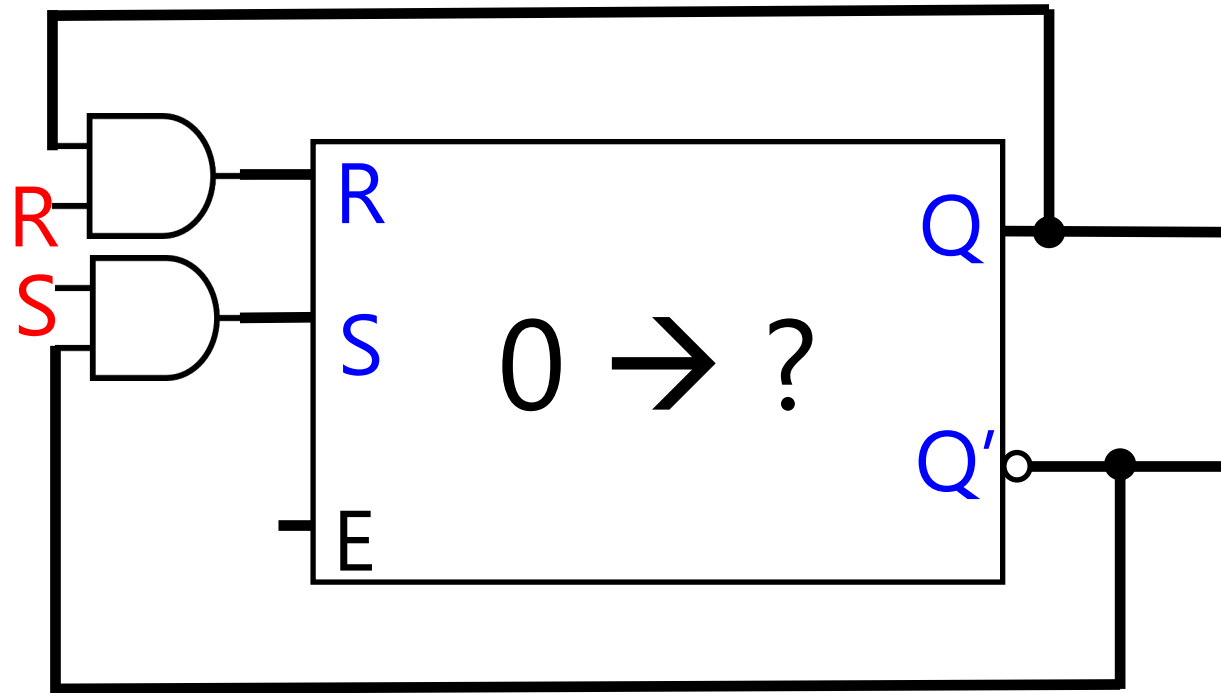
Let's mix T and SR: Store, Set, Reset, **Complement**

Although you have to guess, we'll see a design algorithm for it 😊

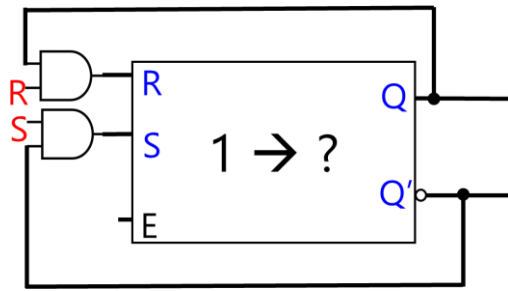




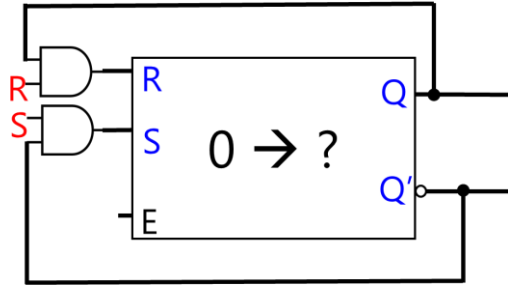
S	R	S	R	Q
0	0	0	0	Store = $Q_t = 1$
0	1	0	1	Reset = 0
1	0	0	0	Store = $Q_t = 1$
1	1	0	1	Reset = 0



S	R	S	R	Q
0	0	0	0	Store = $Q_t = 0$
0	1	0	0	Store = $Q_t = 0$
1	0	1	0	Set = 1
1	1	1	0	Set = 1



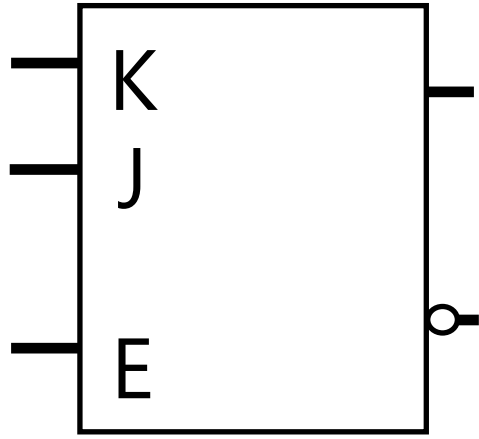
S	R	S	R	Q
0	0	0	0	Store= $Q_t=1$
0	1	0	1	Reset=0
1	0	0	0	Store= $Q_t=1$
1	1	0	1	Reset=0



S	R	S	R	Q
0	0	0	0	Store= $Q_t=0$
0	1	0	0	Store= $Q_t=0$
1	0	1	0	Set=1
1	1	1	0	Set=1

S=J	R=K	Q
0	0	Store= Q_t
0	1	Reset = 0
1	0	Set = 1
1	1	Comp. = Q'_t

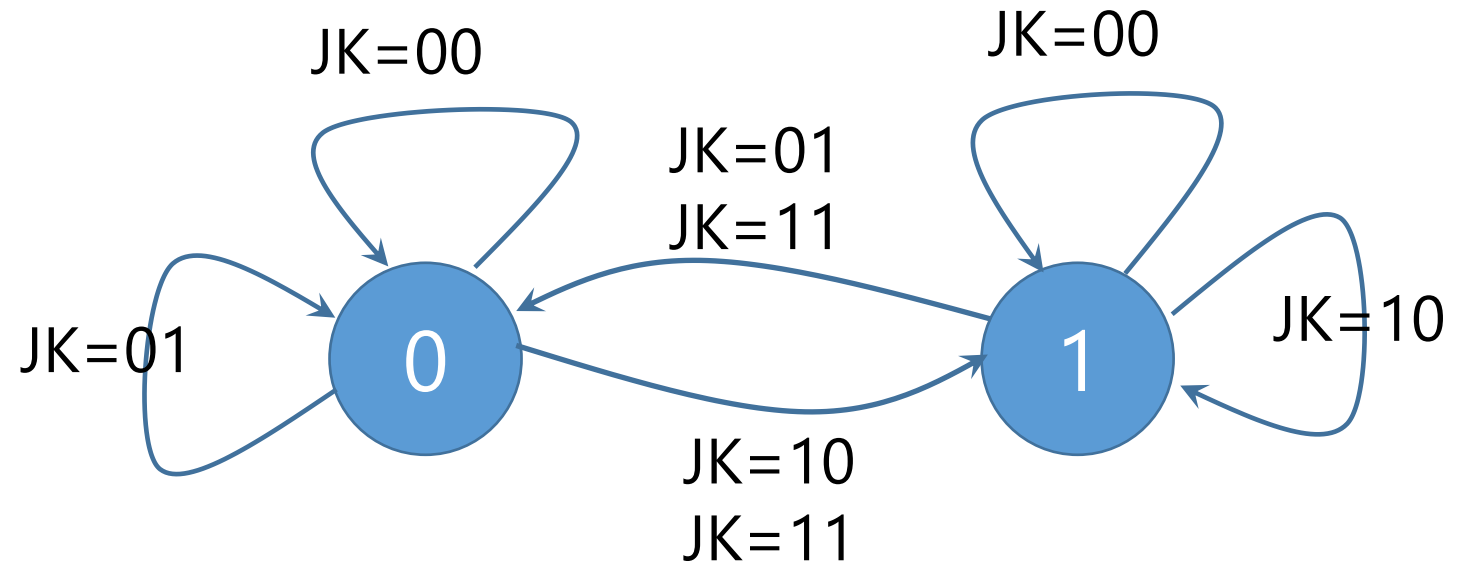
Block Diagram



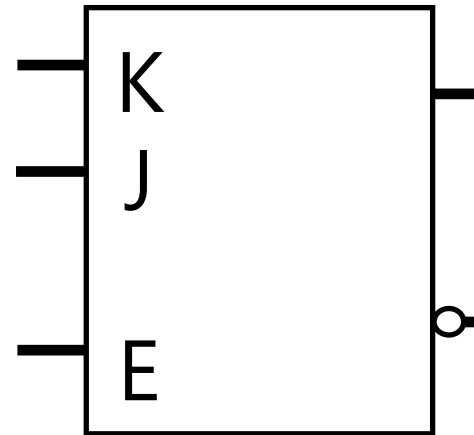
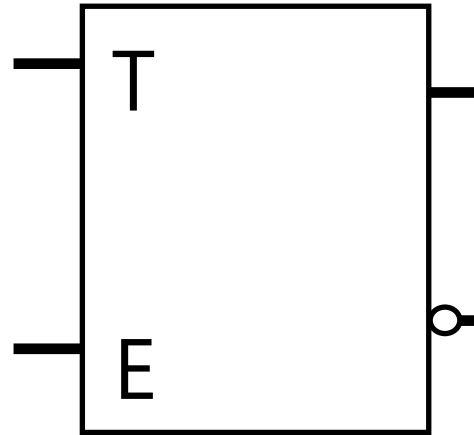
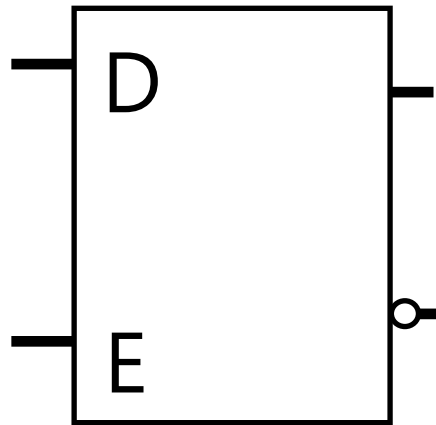
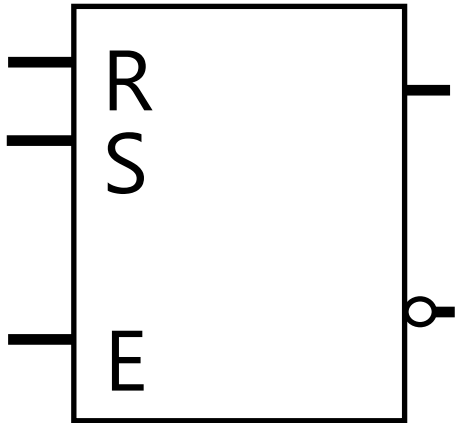
Characteristic Table

J	K	Q
0	0	Q_t
0	1	0
1	0	1
1	1	Q'_t

State Transition Diagram



Recap



S	R	Q
0	0	Q_t
0	1	0
1	0	1
1	1	$\times$

D	Q
0	0
1	1

T	Q
0	Q_t
1	Q'_t

J	K	Q
0	0	Q_t
0	1	0
1	0	1
1	1	Q'_t

ACADEMY AWARD WINNER
ANTHONY HOPKINS

ACADEMY AWARD WINNER
OLIVIA COLMAN

★★★★★
"STUNNING"

"ONE OF HOPKINS'
GREAT SCREEN
PERFORMANCES"

"HOPKINS DELIVERS A
TOUR-DE-FORCE
PERFORMANCE"

"COLMAN IS
DYNAMITE"

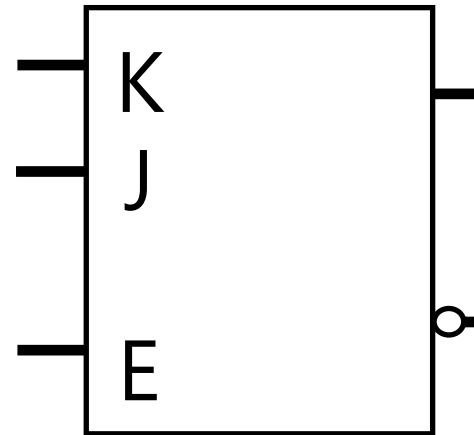
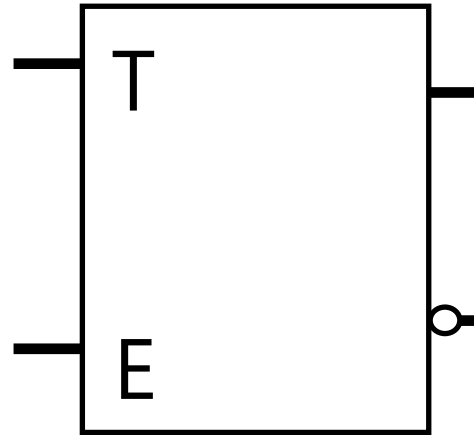
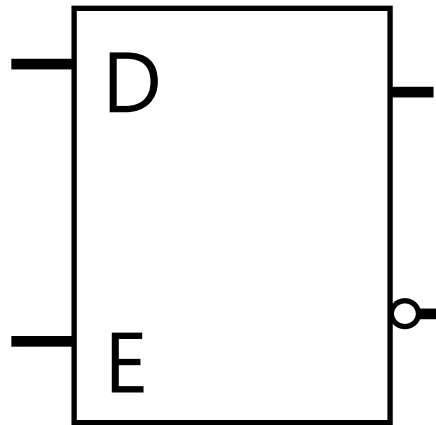
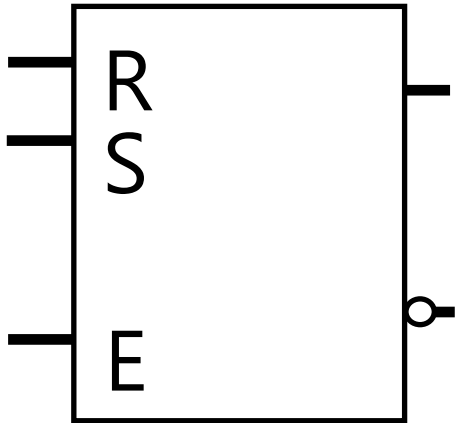
★★★★★
"BREATHTAKING"

MARK GATISS INDOOR FOOTBALL THOMAS SEWELL OLIVIA WILLIAMS

THE



Aerial photographer Brad Walls



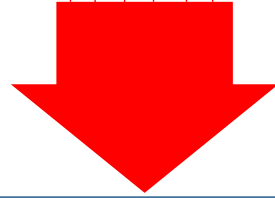
S	R	Q
0	0	Q_t
0	1	0
1	0	1
1	1	$\times$

D	Q
0	0
1	1

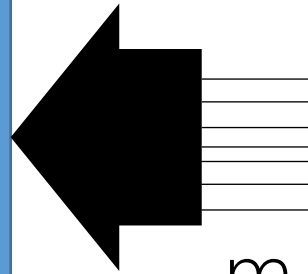
T	Q
0	Q_t
1	Q'_t

J	K	Q
0	0	Q_t
0	1	0
1	0	1
1	1	Q'_t

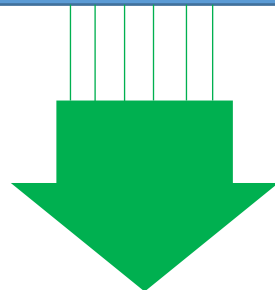
n-bit Data Bus (Write)



$2^m \times n\text{-bit}$
Memory
Module

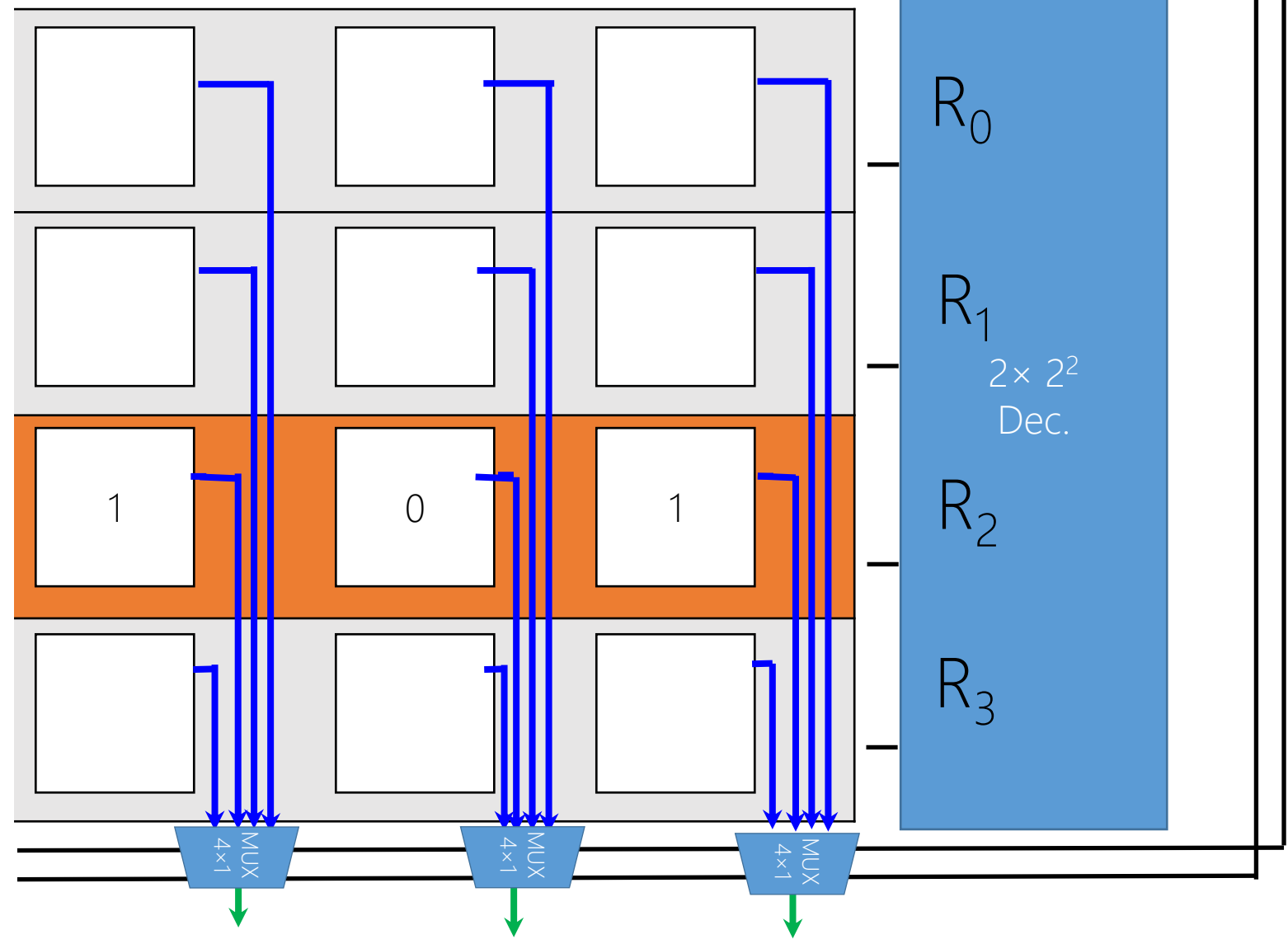


m-bit Address Bus



n-bit Data Bus (Read)

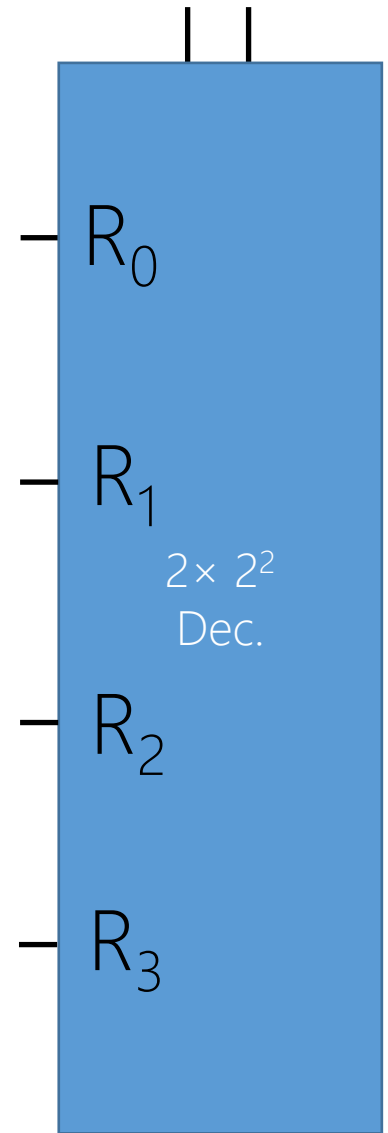
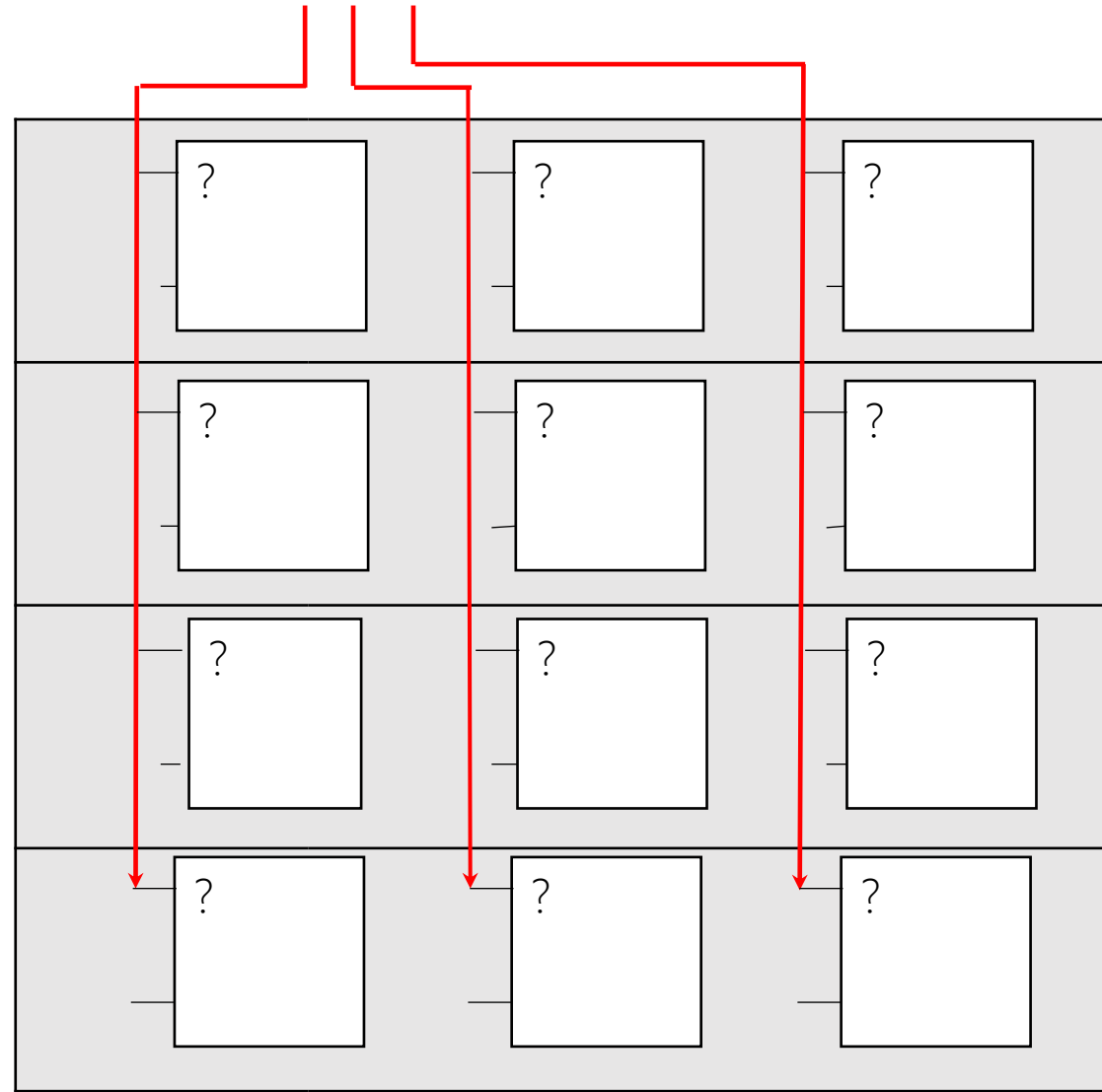
2-bit Address Bus



3-bit Data Bus

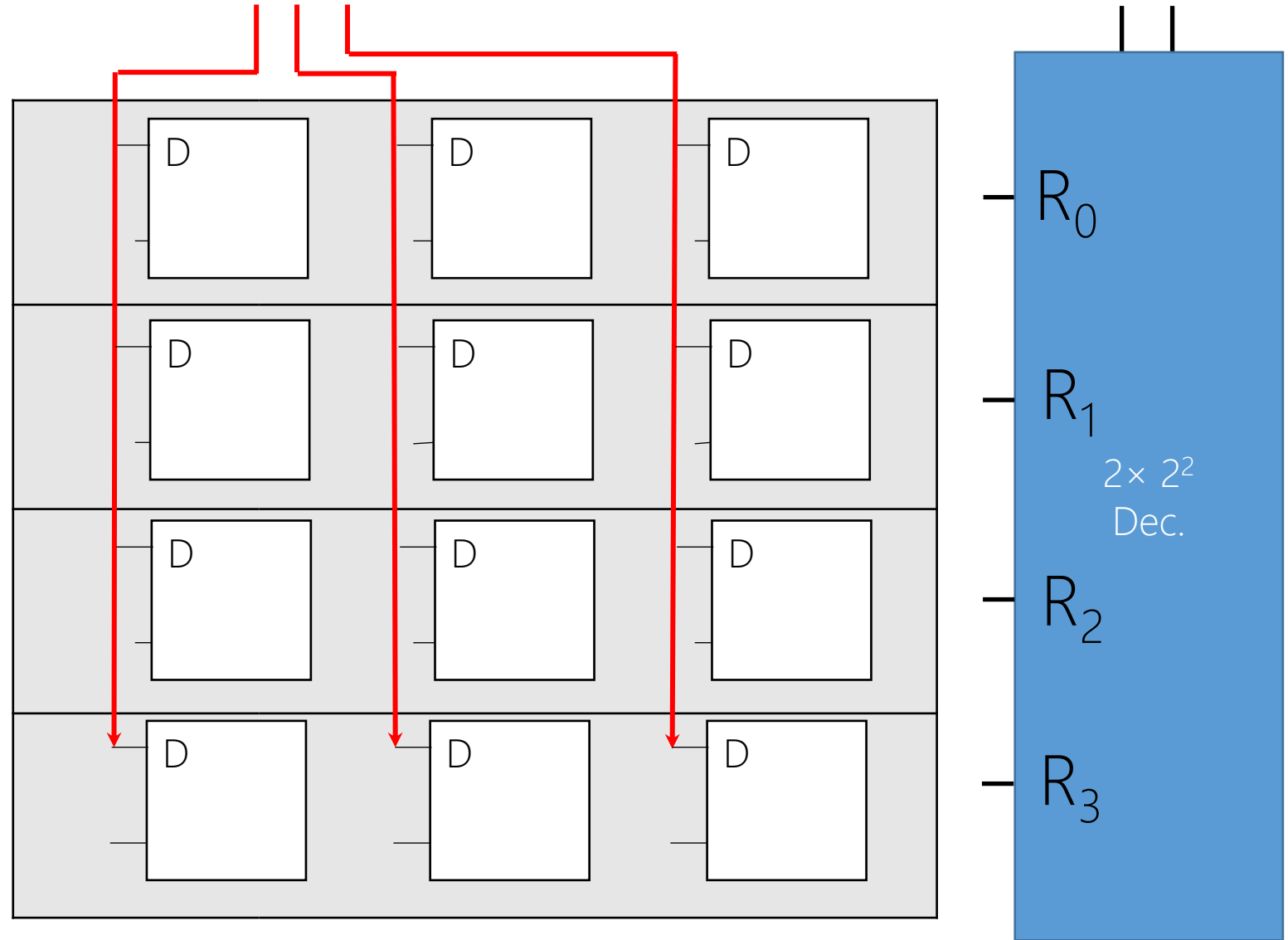
3-bit Data Bus

2-bit Address Bus



3-bit Data Bus

2-bit Address Bus

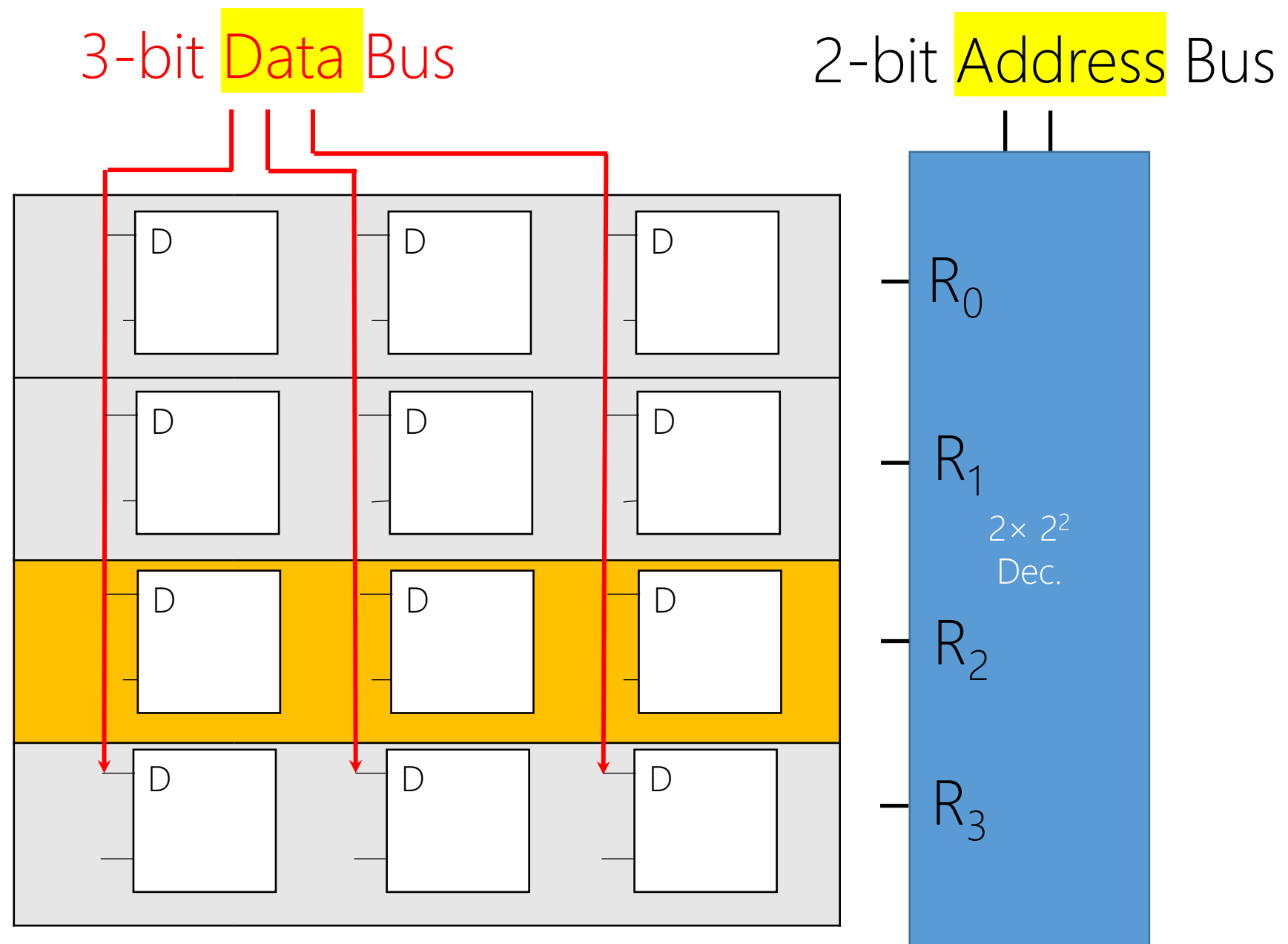


How to direct data
to one row:

Decoder

How to direct data
to all cells of one
row **at the same
time?**

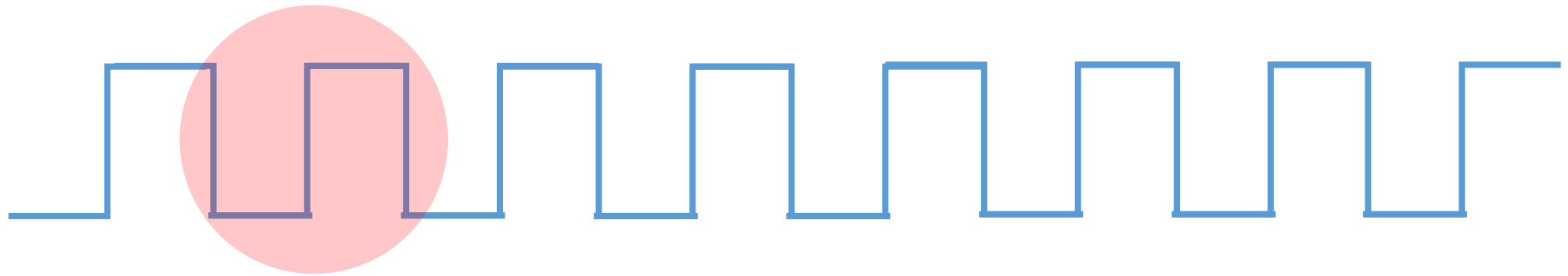
Open their doors
at the same time?!



Clock

shortened as *clk*

timing device that generates a train of pulses



One period is called **pulse**!

Clock

shortened as *clk*

Why we need clock?



Artistic Swimming Olympic Qualifier - Italy is GOLD 🏆

1,355,999 views • Jun 11, 2021

👍 11K 💬 DISLIKE ➦ SHARE ✂ CLIP ⌵+ SAVE ...



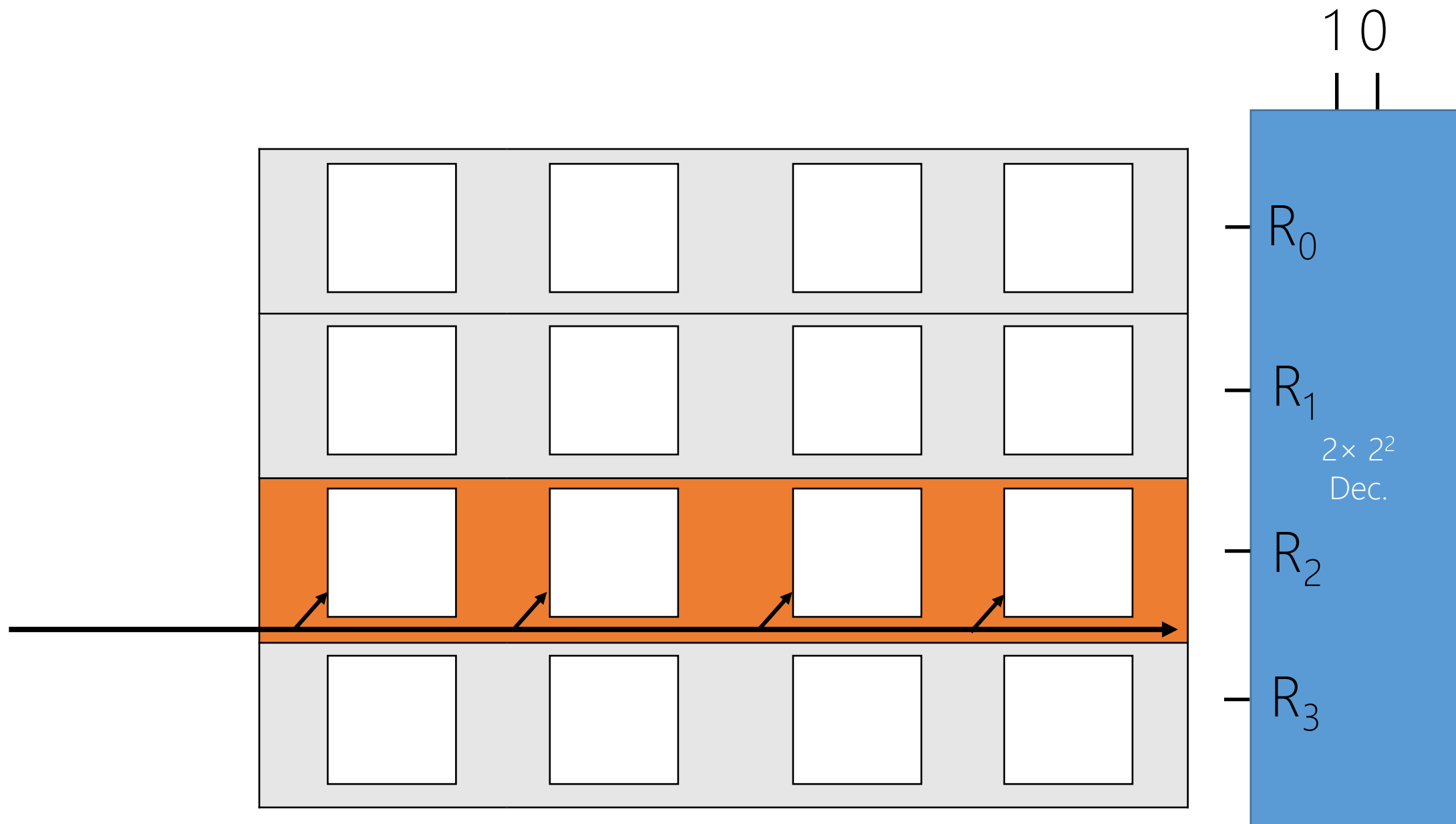
1:17 / 4:45

[#AGT](#) [#AmericasGotTalent](#) [#AGTExtreme](#)

Bonus Performance: Verge Aero Delivers a Gorgeous Drone Show | AGT: Extreme 2022

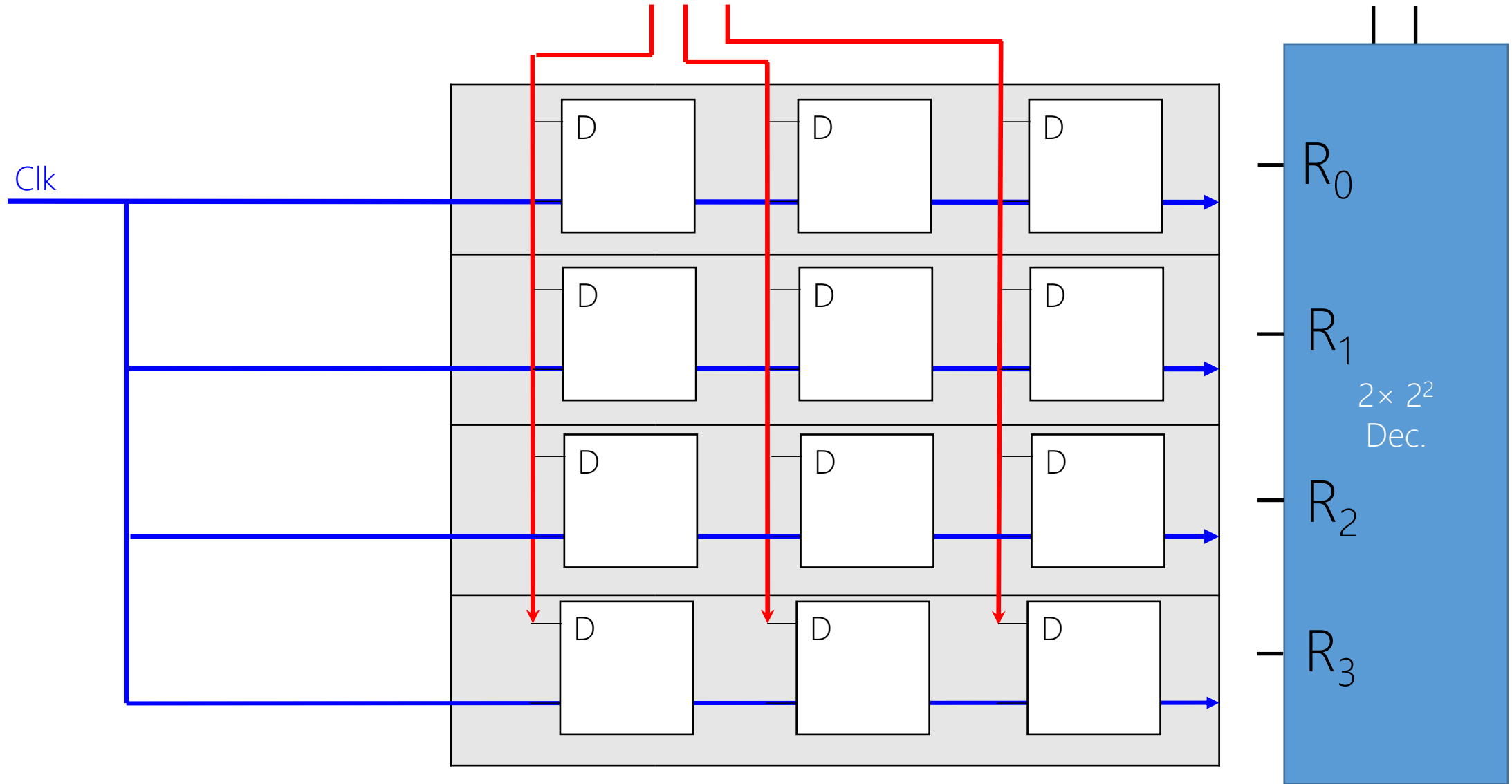
Clock
shortened as *clk*

Synchronize *all* the memory units
when to work



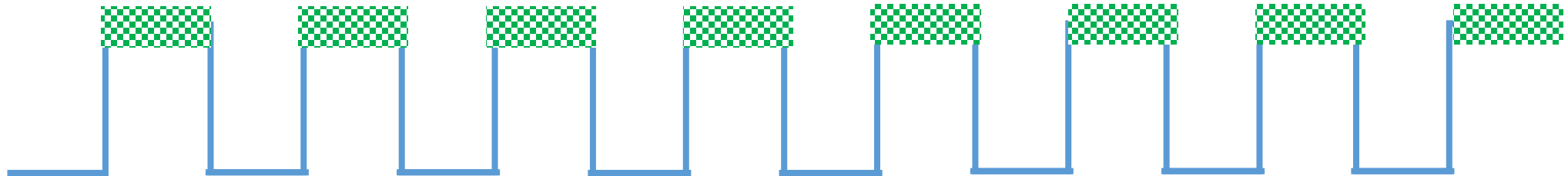
3-bit Data Bus

2-bit Address Bus



Clock

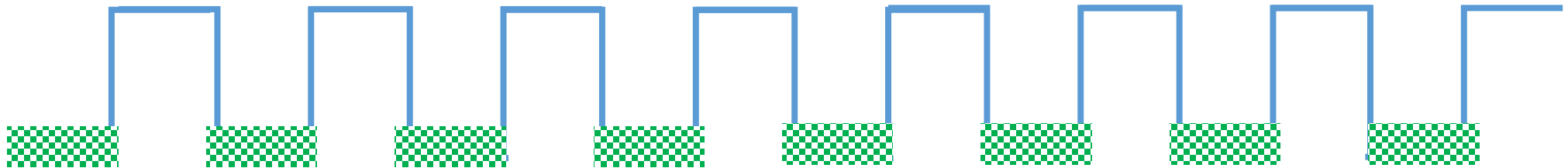
Positive Level (default)

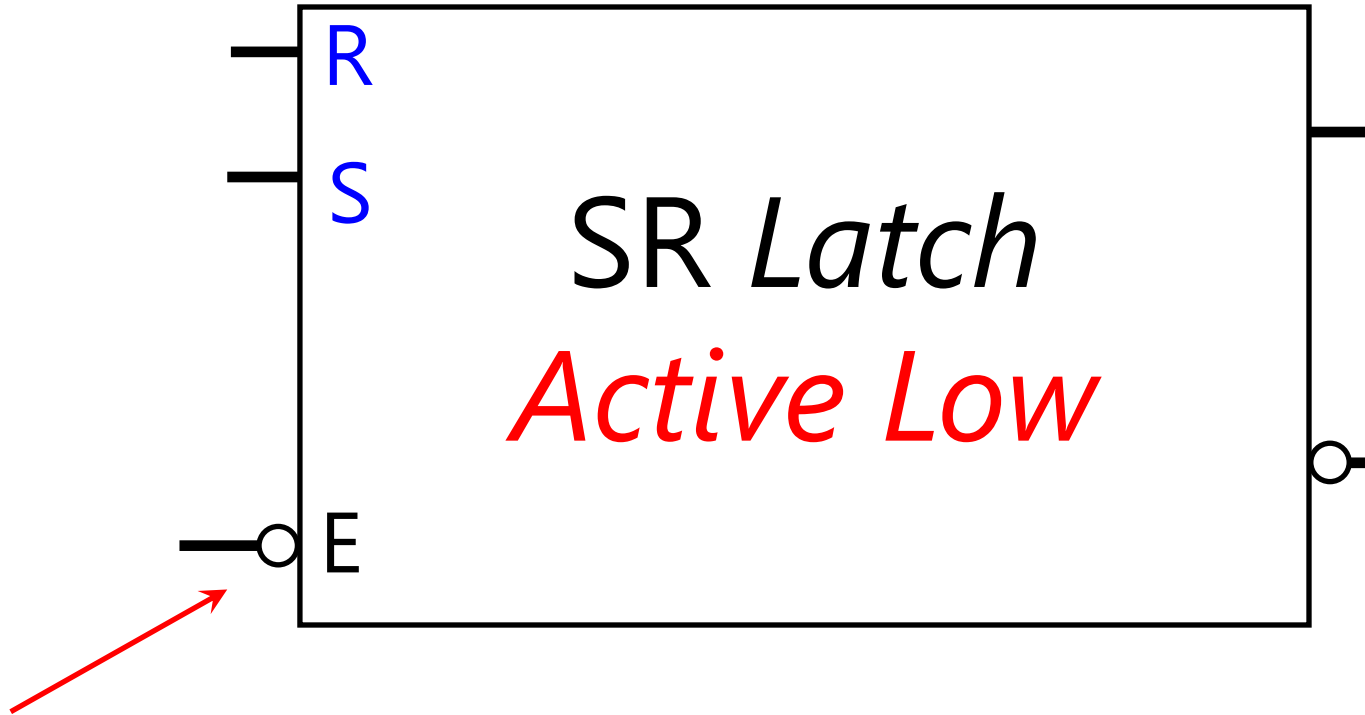




Clock

Negative Level



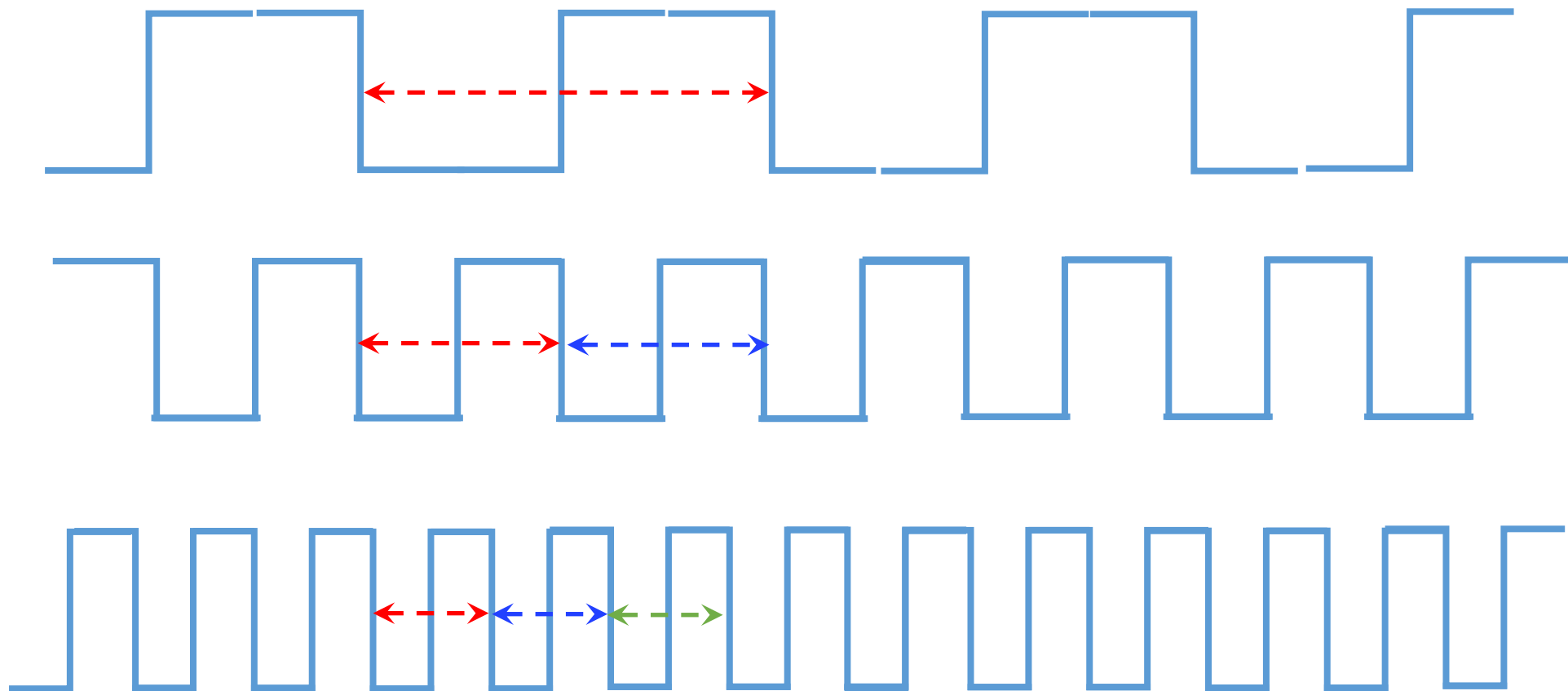


Clock
Frequency (Hz)

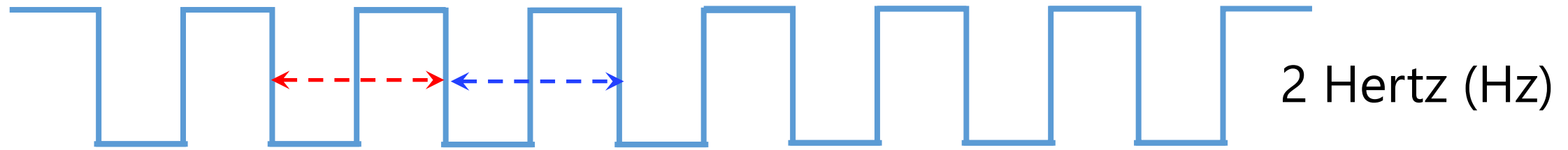
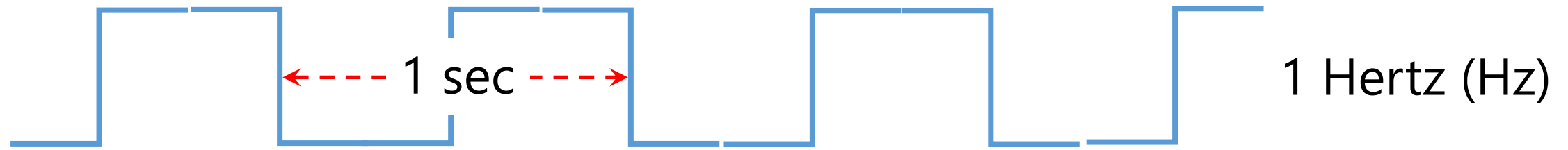


Heinrich Rudolf Hertz

A Metric for Speed



How many pulse in **1** sec?



How many pulse in **1** sec?



How long is one pulse?
 $1/\text{freq. (Hz)}$

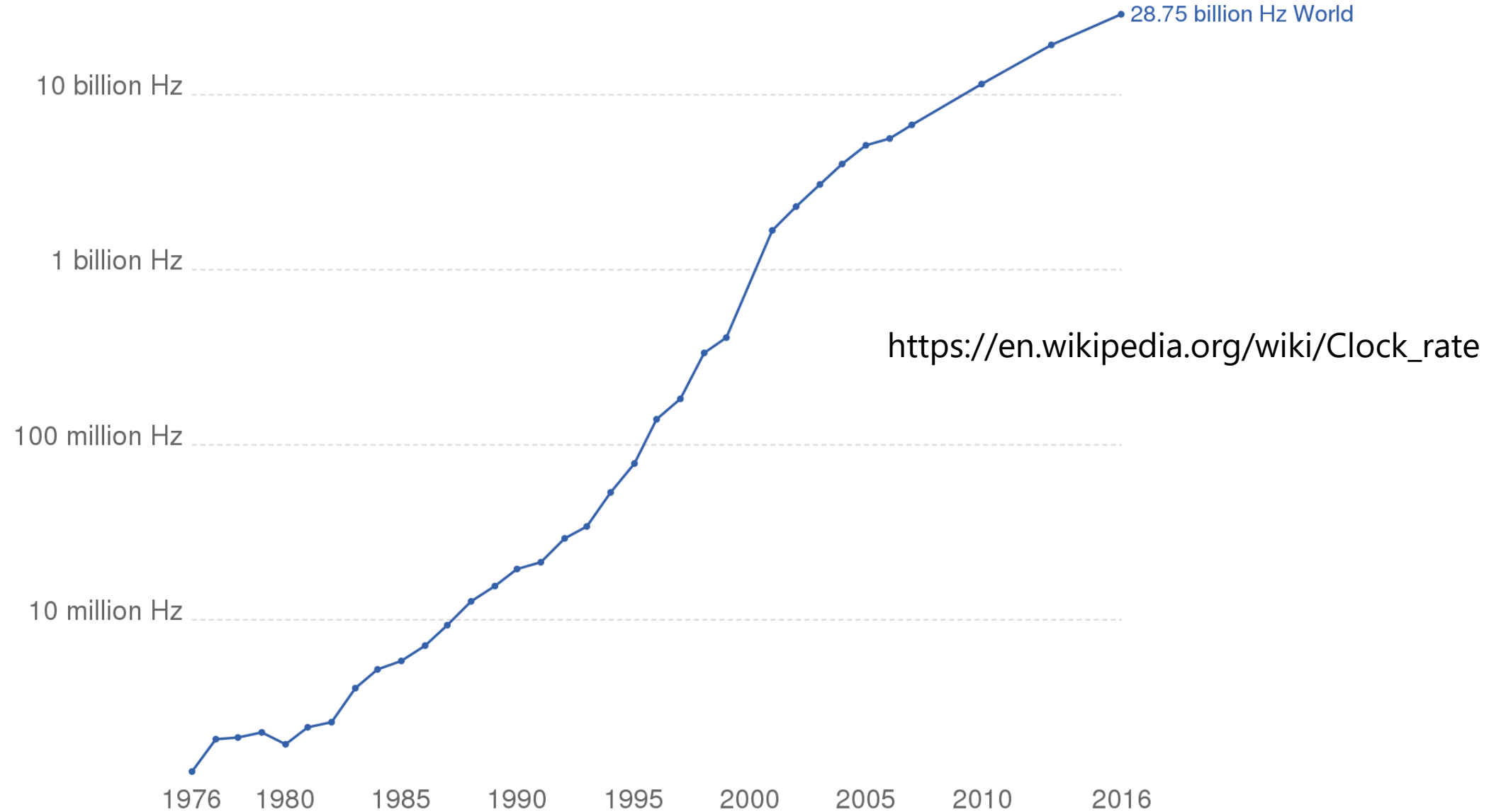


Intel® Xeon® Platinum 8380HL Processor (38.5M Cache, 2.90 GHz)

- 38.5 MB Cache
- 28 Cores 1,000,000,000 (one billion) Hz (hertz)
- 56 Threads 1,000,000,000 (one billion) pulse per sec!
- 4.30 GHz Max Turbo Frequency

Microprocessor clock speed

Microprocessor clock speed measures the number of pulses per second generated by an oscillator that sets the tempo for the processor. It is measured in hertz (pulses per second).



Source: Ray Kurzweil (2005, updated to 2016). The Singularity Is Near: When Humans Transcend Biology.

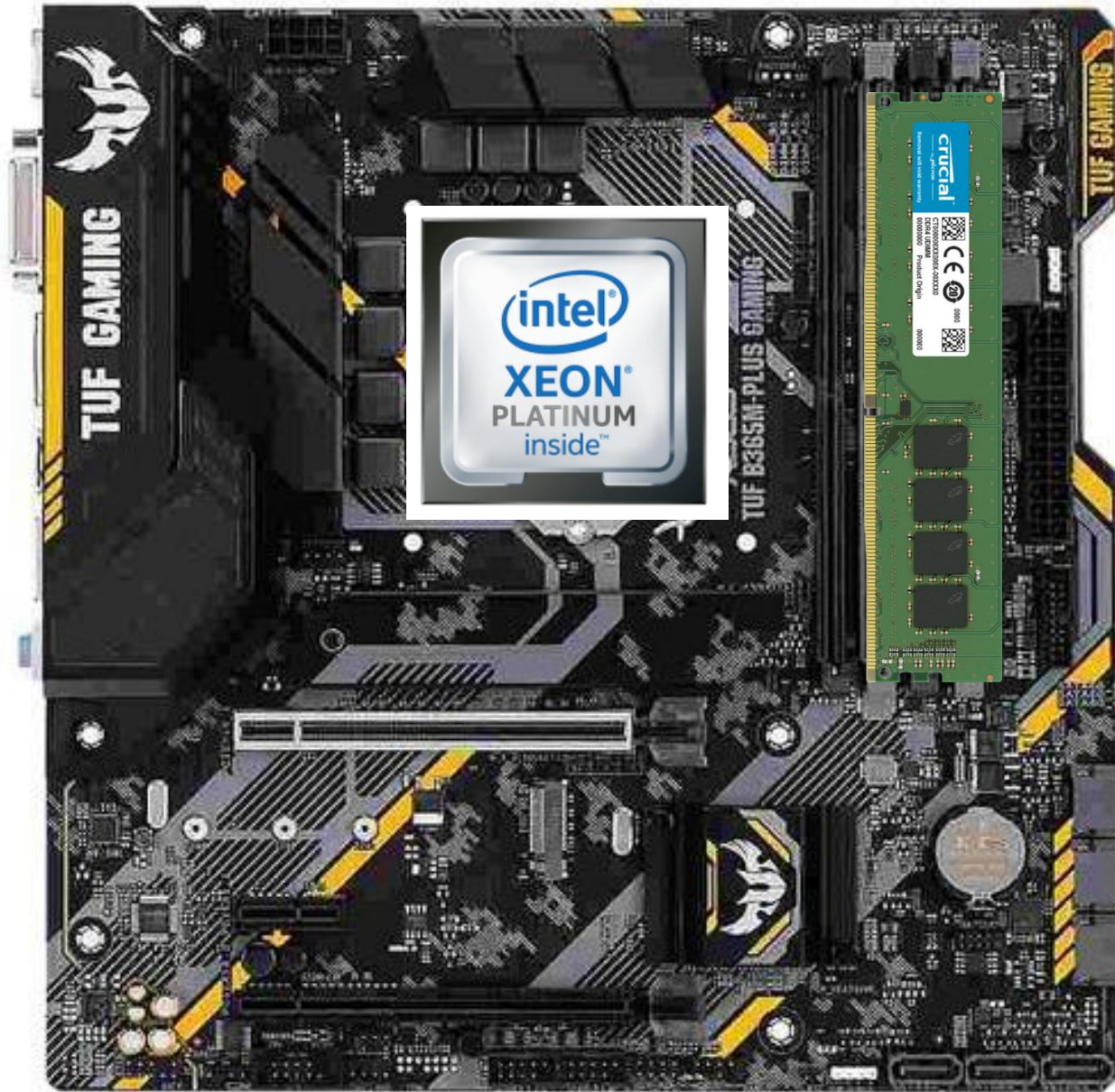


Speed - DDR4-2666 ⓘ

Warranty - Manufact

CAS latency - 19 ⓘ

The number of transfers per clock cycle times the clock frequency, expressed as MegaTransfers per second.

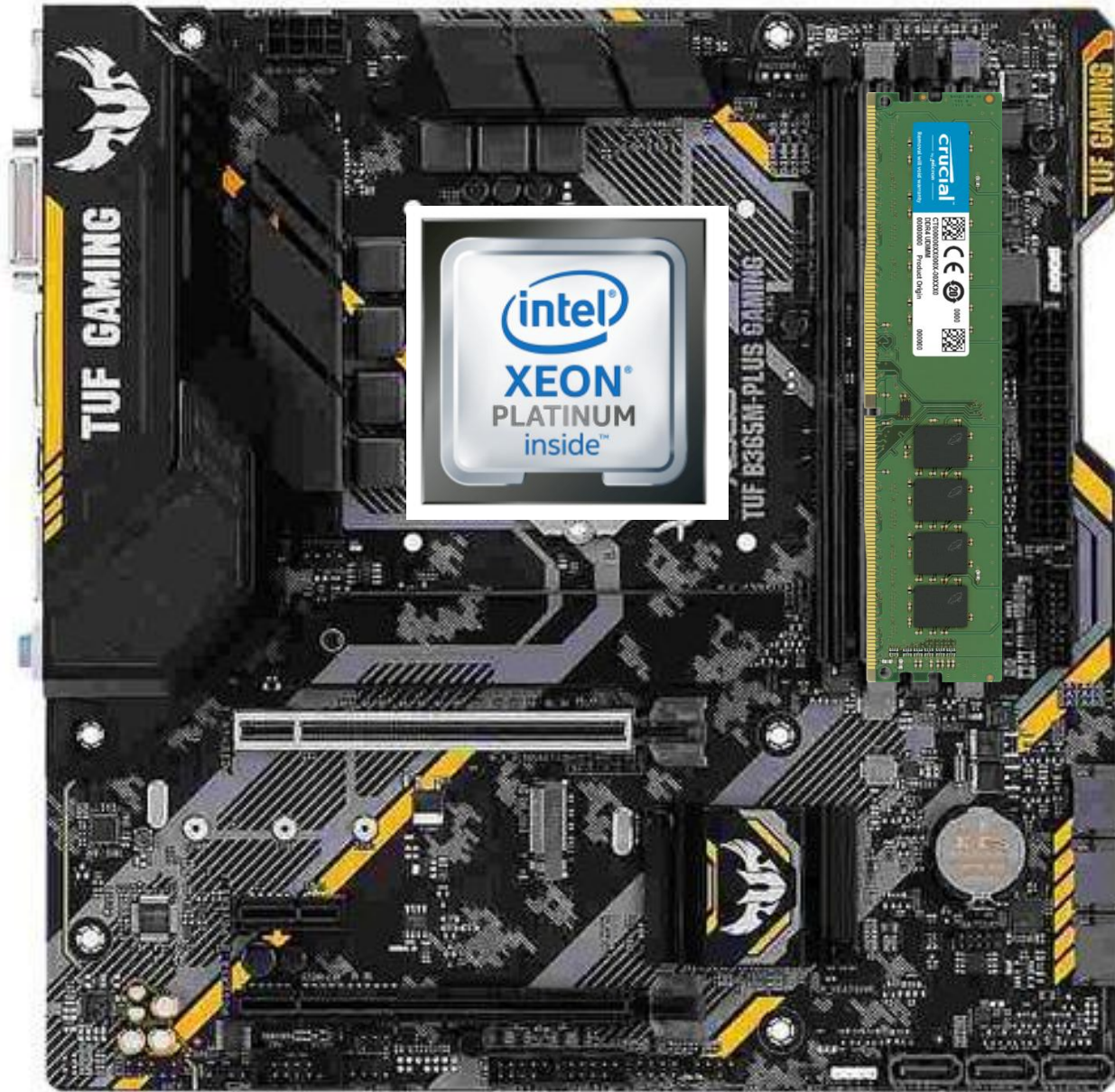


CPU: X Hz

Memory: Y Hz

Mainboard (BUS): Z Hz

Final Speed?



CPU: X Hz

Memory: Y Hz

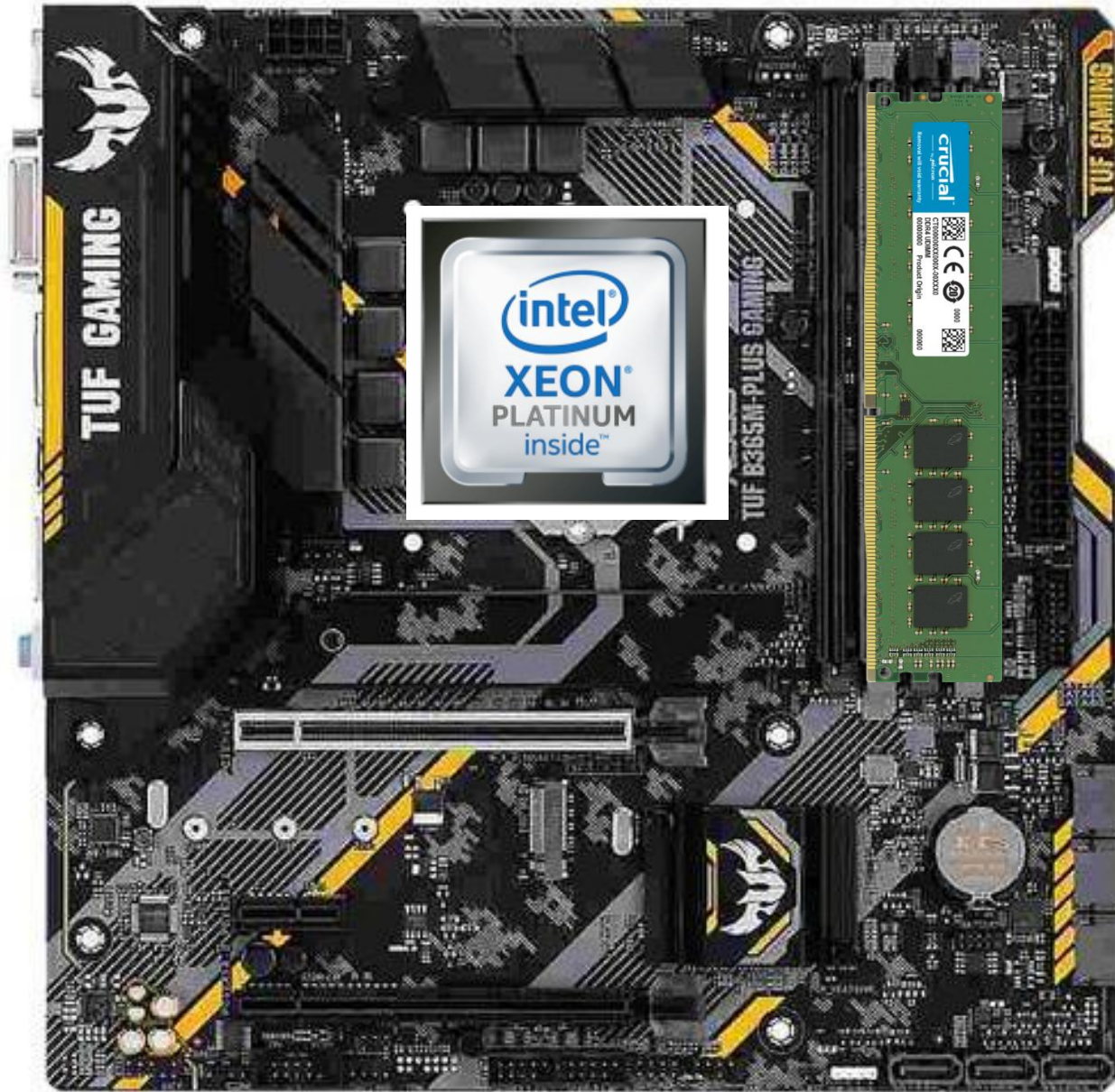
Mainboard (BUS): Z Hz

At market:

$$X > Y = Z$$

$$X = 2.9\text{GHz}$$

$$Y = Z = 2.6\text{GHz}$$



CPU: X Hz

Memory: Y Hz

Mainboard (BUS): Z Hz

Final Speed:

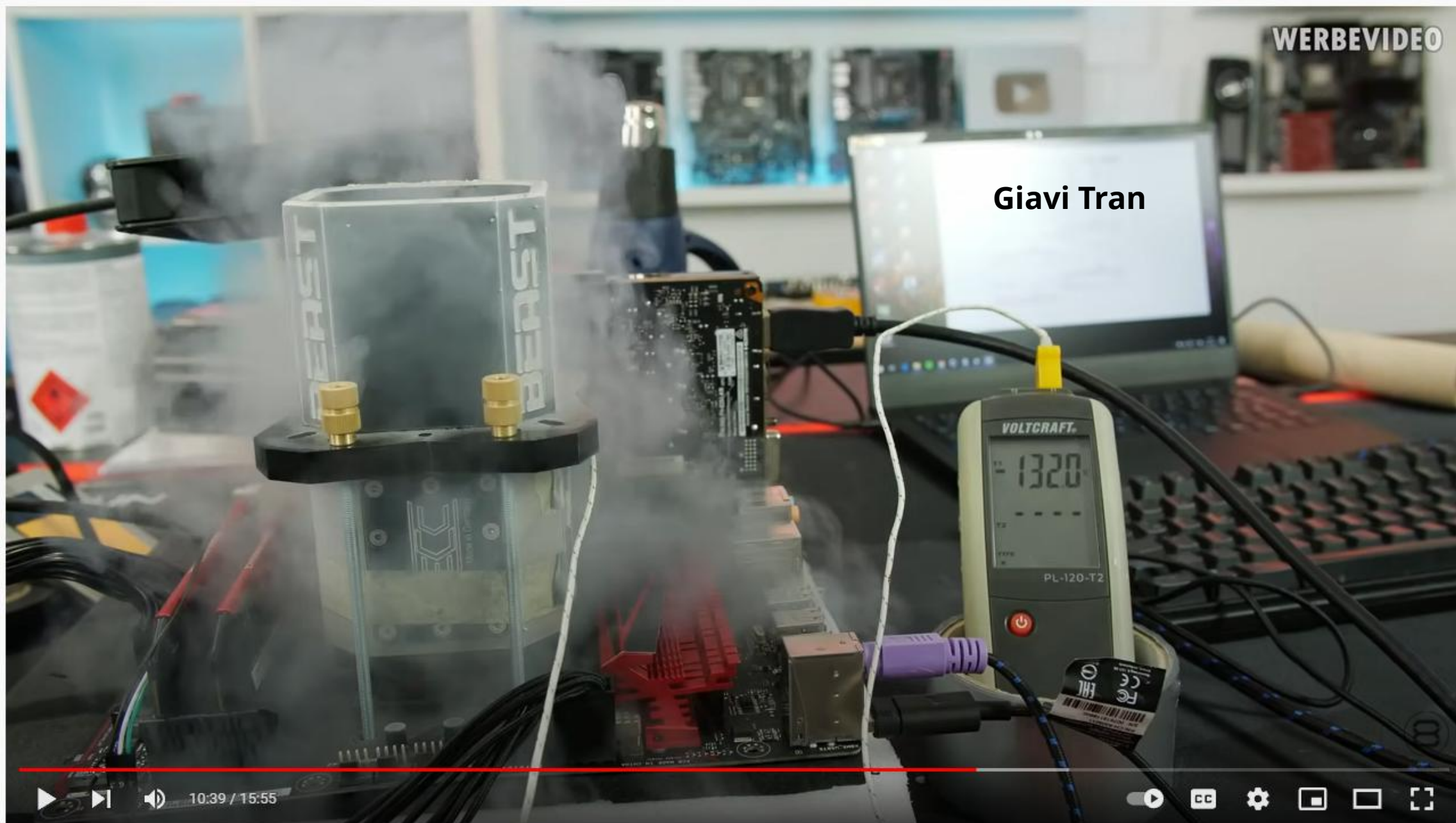
CPU internal: X

CPU external $\leftrightarrow$ Memory

$Y=Z= 2.6\text{GHz}$

Overclock?

Giavi Tran



We Overclocked an AMD CPU to over 8.1 GHz !!

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